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Upper Cretaceous–Paleocene magnetic stratigraphy at Gubbio, Italy IV. Upper Maastrichtian-Paleocene Magnetic Stratigraphy

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Upper Cretaceous–Paleocene magnetic stratigraphy at Gubbio, Italy I. Lithostratigraphy and sedimentology

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ABSTRACT

An essentially complete section of middle Cretaceous to Paleocene, calcareous, pelagic sediments is exposed at Gubbio, in the Umbrian Apennines of Italy. This, the first of five papers, presents the lithostratigraphic and sedimentological basis for subsequent discussions of biostratigraphy and magnetic stratigraphy. Biogenic coccolith-globigerinid oozes were deposited along with a constant but low background of fine terrigenous detritus at a mean sedimentation rate (compacted) of 6.6 m/m.y. from Albian to Eocene time; individual stages deviate from this by a factor of two. Short-term variations in carbonate input yielded a sequence of limestone beds punctuated by thin shale interbeds. These shale-limestone couplets accumulated in time spans between 20,000 and 100,000 yr, and the thin shales may represent up to one-half of the total time. Modification by diagenetic solution, tectonic shear, and weathering render shale interbeds unsuitable for magnetic work. The limestone beds, first bioturbated and then compacted and cemented during diagenesis, appear likely to have recorded magnetic history at intervals on the order of 50,000 yr, sufficient to provide a good record of polarity changes. Moderate to high states of oxidation prevailed except in Aptian-Albian and a portion of Turonian to Santonian times.

INTRODUCTION

This is the first of a series of five papers dealing with the middle Cretaceous to Paleocene sediments exposed in the Bottaccione gorge near Gubbio, Italy (Alvarez and others, this issue, Fig. 1; and Carta Geologica d'Italia, sheet 116, "Gubbio"). The various authors have been carrying out an integrated study of lithostratigraphy, biostratigraphy, sedimentology, geochemistry, and paleomagnetism. This paper presents the measured section and preliminary results of sedimentological-geochemical investigations still in progress — material necessary as a base for the subsequent papers.

LOCATION, PREVIOUS WORK, AND GEOLOGIC SETTING

The Bottaccione section, located within a kilometre of the town of Gubbio in the Umbrian Apennines, promises to be of extraordinary importance for the understanding of the Cretaceous and Paleogene. Renz (1936, 1951) initially called attention to this apparently unbroken sequence of pelagic sediments, mainly limestone, of Tithonian or Neocomian through Eocene age. Studies by Luterbacher and Premoli Silva (1962, 1964) showed the presence of all standard Paleocene and Eocene foraminiferal zones, including a new, earliest Paleocene zone — that of *Globigerina eugubina*, now recognized at several other localities. Cretaceous foraminiferal zones down to that of *G. concavata* have also been recognized (Premoli Silva, this issue). The geology of the Gubbio area was described by Barnaba (1958), and preliminary data on the paleomagnetism of the Gubbio sequence were presented by Alvarez and Lowrie (1974), Lowrie and Alvarez (1974, 1975), and Premoli Silva and others (1974).

The Umbrian Apennines are probably parautochthonous, lying to the east of the allochthonous units of the northern Apennines (see Bortolotti and others, 1970, for general relations). As in certain other Tethyan areas, such as the southern Alps, depositional environments in the Umbrian region evolved from a carbonate shelf setting in Triassic-Liassic time into a pelagic phase. This, commencing in the Jurassic and extending into the Paleogene, was succeeded by Miocene terrigenous turbidites. Main structural features in the Umbrian ranges are large anticlines, trending northwest, broken by normal faults. The Bottaccione gorge cuts through the northeastern limb of such a faulted anticline; bedding strikes average N50°W and northeastern dips range from 35° to 55°. Slump structures or intrastratal folds are visible on the northwestern wall of the gorge and in the Valle della Contessa farther west, but are insignificant on the southeastern side of the gorge where our section was measured and samples taken. Road cuts provide almost continuous outcrops of comparatively fresh rock, and occasional gaps can generally be filled by outcrops higher on the hillside, especially along the Medieval aqueduct (the Bottaccione).

The Cretaceous-Paleocene rocks have been buried to a depth of 1 to 2 km under Miocene flysch. Limestones lack porosity and break with a smooth conchoidal fracture: bedding-parallel and bedding-oblique stylolites are common, but no pervasive recrystallization is evident, and fossils are undistorted. Shales and marlstones, though compacted and fissile, show no tectonic cleavage. The region is free from intrusives and evidence of thermal alteration.

LITHOSTRATIGRAPHY

All paleomagnetic samples, biostratigraphic zones, and other points of observation in the succeeding four papers refer to a section measured by the authors. This commences at the contact of the massive, light-gray Maiolica Limestone, of Barremian-Aptian age, with dark, fissile shales and marls of the Aptian-Albian Scisti a Fuciodi. It extends from more than 400m, into Eocene rocks. A generalized version of this section is shown in Figure 1, along with the stage and zonal boundaries provided by Premoli Silva.

Traditionally, this part of the section has been divided into three main lithic units or formations: the Scisti a Fucoidi, the Scaglia (pronounced Scáhl-ya) Bianca, and the Scaglia Rossa. The distinctive characteristics of these units are described below, and certain chemical and mineralogical aspects of the major rock types are shown in Table 1.

Scisti a Fucoidi

This unit extends from 0 to 47 m in the measured section. It is composed primarily of shales and marlstones that show wide varia-

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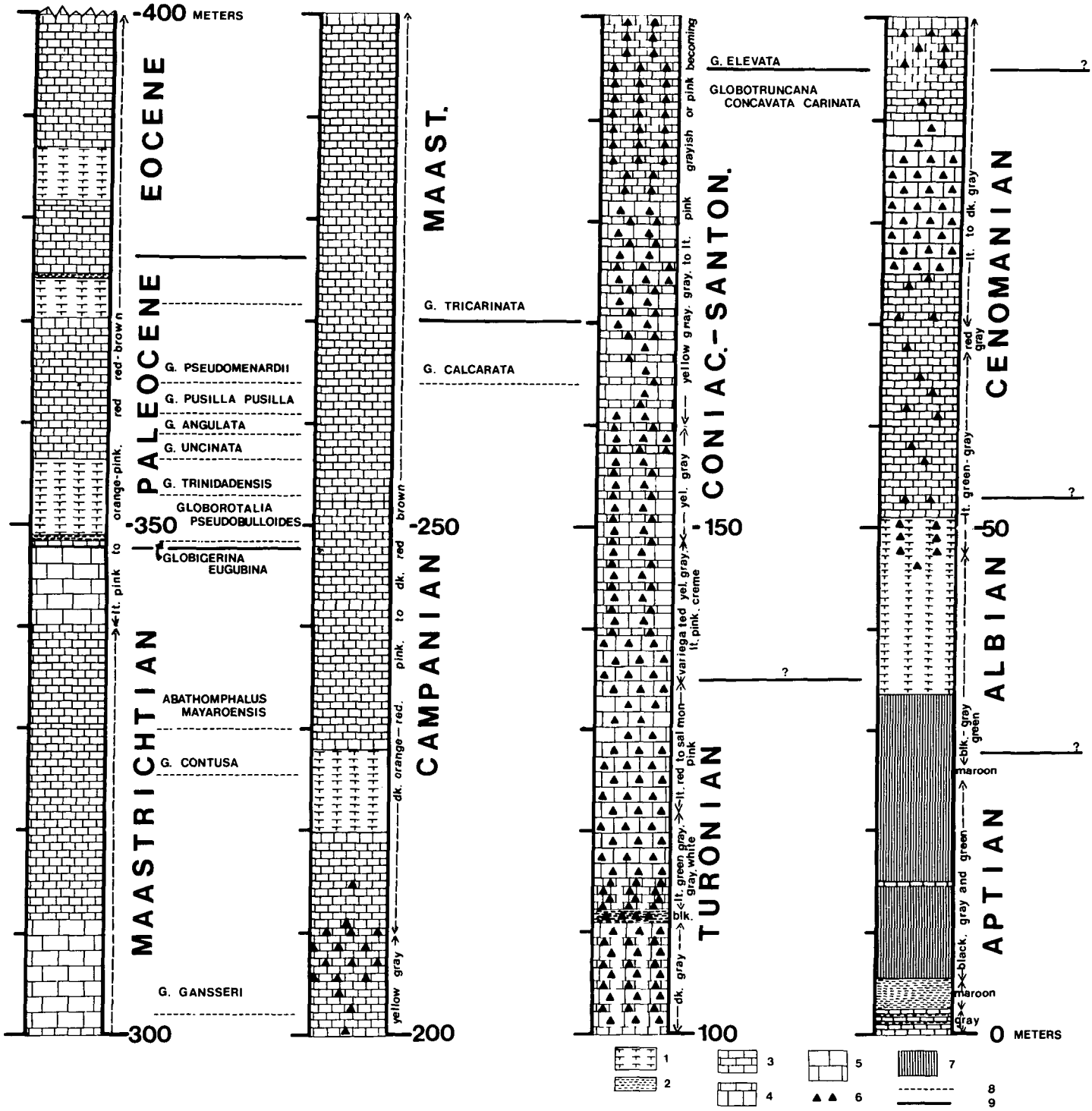


Figure 1. Generalized stratigraphic section, Gubbio. Planktic foraminiferal zonation and Cretaceous stage boundaries by Premoli Silva (this issue). Boundaries for Aptian-Albian, Albian-Cenomanian, Cenomanian-Turonian, and Turonian-Coniacian are provisional.

General rock colors shown along columns give an indication of oxidation state (see text). Symbols: 1. marlstone and shaly marlstone; 2. shale; 3. thin-bedded limestone (<25-cm beds); 4. variably bedded limestone; 5. thick-bedded limestone (>25-cm beds); 6. chert: density of symbols represents abundance; 7. rhythmically bedded shales and marls, black to light-green and gray colors alternate; 8. foraminiferal zone boundaries; 9. Stage boundaries (epochs in Cenozoic). Formations: Scisti a Fucoidi (0 to 47 m); Scaglia Bianca (47 to 115 m); Scaglia Rossa (115 to 440 m).

tion in oxidation state, ranging from laminated, black, organic carbon-rich shales to light-green and, more rarely, red shales and marlstones. Bioturbation is evident in the more oxidized portions. Rhythmic alternations of black shale and green marlstone ranging in thickness from 30 to 110 cm are characteristic. Calcium carbonate content ranges from 20 to 70%, generally increasing toward the gradational boundary with the Scaglia Bianca. Quartz, mica-illite, pyrite, chlorite, and some montmorillonite are also common. High iron values occur in pyrite-rich beds.

Scaglia Bianca

The lower and upper boundaries of this unit are placed at 47 and 115 m. It consists predominantly of siliceous, gray to light-green

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TABLE 1. CALCIUM CARBONATE, ORGANIC CARBON, AND SEDIMENTARY IRON CONTENTS OF MAIN LITHOLOGIES IN GUBBIO SECTION

Lithology	Range in weight percent		
	$CaCO_3$	Org. C	Iron as Fe_2O_3
Black shales and marlstones	26.0–60.0	0.25–13.6	1.28–3.45
Gray to green shale and marlstones	26.8–64.6	0.07–0.25	1.54–3.83
Gray to green limestone and marly limestone	65.0–92.0	0.01–0.17	0.164–2.79
Maroon shale	39.3–46.4	0.08–0.09	3.10–5.02
Pink to red shale and marlstone	23.0–65.0	0.02–0.17	0.615–9.15
Pink to red limestone and marly limestone	65.0–92.3	0.01–0.07	0.232–0.888

limestones, with a few pink to red beds. Calcium carbonate content is generally higher than in the underlying Scisti, and organic carbon content lower, with the exception of the "Bonarelli bed" (Scisti ittiolitici) at 111 m, a 1-m-thick cherty black shale with as much as 13.6 percent organic carbon. Nodular cherts, usually greenish-gray but in some cases red-brown, occur at intervals ranging from 15 to 50 cm. The uppermost limestone beds, grading into the Scaglia Rossa, alternate between gray and pink colors. Burrow-mottling is common, especially in zones of contrasting lithologies; however, unburrowed laminated limestones do occur. Bedding thicknesses range from 10 to 110 cm, averaging about 25 cm, with thinner shale interbeds of generally less than 2 to 3 cm. Mineralogy is similar to that of the Scisti a Fucoidi.

Scaglia Rossa

This unit extends from 115 to 440 m in the measured section. It is predominantly red, homogeneous limestone and marly limestone that are generally more than one-half calcium carbonate except in shale interbeds which define much of the bedding. Organic carbon content is extremely low, except in some gray to yellowish-gray, partly laminated to burrow-mottled, somewhat pyritic limestones that occur in the Santonian-Coniacian portion. Red to gray nodular cherts occur in the lowermost, pre-Campanian interval, and red nodular chert recurs in the early middle Eocene limestone. Bedding thicknesses vary from 15 to 85 cm, again averaging about 20 cm. Quartz, mica-illite, and some montmorillonite are the common noncarbonate minerals, accompanied by traces of hematite and magnetite. Pyrite is present in some non-red limestones in the lower part of the unit.

SEDIMENTATION

The sediments of the Gubbio sequence consist primarily of biogenic calcite, mainly coccoliths and foraminifera, and a fraction of terrigenous clay and silt which is possibly wind-blown. Problems which particularly interest us are those of sedimentation rates, of the origin of rhythmic bedding, of burrow-mixing, and of variations in oxidation state. All of these are interrelated to some degree and are of concern to paleomagnetic studies.

Sedimentation Rates

The overall mean sedimentation rate for the Albian through Paleocene sequence (for compacted rock) is 6.6 B (Bubnoff units; Fischer, 1969; μ/yr., mm/thousand yr., m/m.y.), using the figures of 349 m sedimented in 52.5 m.y. (Table 2). Individual stages show rates differing by no more than a factor of two from the average, ranging from 2 to 3 B in shales of the Albian and Paleocene to 8 to 13 B in more calcareous Upper Cretaceous units.

Very similar results may be obtained on a finer scale by a somewhat independent approach, which employs an interplay between sedimentological and magnetic studies (Arthur and Roggenthen, 1976). If the Gubbio magnetic polarity zones for the Maastrichtian-Paleocene are properly correlated with sea-floor magnetic anomalies (Roggenthen and Napoleone, this issue), and if we assume that sea-floor spreading through this time interval was essentially continuous at constant rate, then the stratigraphic section may be subdivided into time units proportional to the width of corresponding sea-floor anomalies bracketed by acceptable stratigraphic dates (Heirtzler and others, 1968, with corrections by Sclater and others, 1974). Such calculations yield mean sedimentation rates of 8 to 13 B for polarity zones of the calcareous upper Maastrichtian and values of 2 to 5 B for those of the shalier Paleocene.

TABLE 2. SEDIMENTATION RATES, GUBBIO SEQUENCE

Epoch or age	Thickness (m)	Absolute age*	Sedimentation rate (m/m.y.)	Absolute age†	Sedimentation rate (m/m.y.)	Absolute age§	Sedimentation rate (m/m.y.)
Paleocene		53.5		53.5		53.5	
	27.4		2.9		2.4		2.4
		63.0		65.0		64–65	
Maastrichtian	77.4		9.7		15.5		12.9
		71.0		70.0		70–71	
Campanian	70.2		7.8		11.7		6.4
		80.0		76.0		82±	
Coniacian-Santonian	65		10.8		6.5		13.0
		86.0		86.0		87±	
Turonian	40		8.0		6.7		13.3
		91.0		92.0		89–90	
Cenomanian	42		10.5		5.3		10.5
		95.0		100.0		94±	
Albian	25		2.3		4.2		2.1
		106.0		106.0		106(?)	

Note: Rates calculated using three different radiometric age schemes.
* R. Douglas and D. Bukry (1974).
† van Hinte (1972); Berggren (1972).
§ Obradovich and Cobban (1975).

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A wider range in sedimentation rates exists at the level of individual beds, to be considered next.

Bedding Rhythms

The middle and upper parts of the sequence are characterized by rhythmic alternation of more massive and generally bioturbated limestones, and marlstone or shale interbeds of lesser thickness. In the lower part (Scisti a Fucoidi), this alternation is one of bioturbated, massive, light-colored claystones or marlstones, and dark, fissile, laminated shales; this intercalation passes upward into the Scaglia where it becomes the more typical thick, bioturbated limestone beds and generally thin fissile shale or marlstone partings. Throughout the section, massive beds range in thickness from 10 to 85 cm, averaging between 20 and 25 cm; in more rapidly deposited parts of the sequence such as the Campanian, the massive beds are generally thicker than in the more slowly deposited portions such as the Paleocene. The shale interbeds are generally an order of magnitude thinner, ranging from millimetres to several centimetres in thickness. Number of bedding couplets per time interval indicates a duration of between 20,000 and 100,000 yr per couplet (Arthur and Roggenthen, 1976).

In theory, these bedding couplets could have arisen either by alternating episodes of sedimentation, or as a result of diagenetic dissolution of carbonate along rhythmically spaced, bedding-parallel planes. We believe that they represent a compromise between these alternatives.

Burrows in the purer limestones are not uncommonly filled with the darker, shalier matrix characteristic of the shale or marl interbeds. Hence, at least a portion of the present bedding structure must be of primary origin.

On the other hand, the tightly cemented nature of the limestone beds and the absence of crushing or distortion of foraminifera within them show that the limestones were lithified primarily by cementation, rather than by compaction. The abundance of stylolites and non-digitate solution planes near the upper and lower margins of limestone beds demonstrates that solution was active within the section. Profiles of calcium carbonate content across the couplets show highest values in the mid-portion of limestone beds. Together, these observations suggest that there has been a transfer of carbonate from the shaly interbeds and the margins of limestone beds to their middle.

Accordingly, we believe that the present bedding character resulted from an original alternation of carbonate-rich and carbonate-poor episodes, and that it was subsequently modified, first of all by burrowing which tended to blur the initial contrast, and then by interstratal solution transfer, which tended to re-emphasize it. In addition, the shaly beds have subsequently undergone some shear in the course of folding, and have been affected by differential weathering.

The initial variation, from a more calcareous to a more clay-rich phase, is most easily explained by the assumption that one phase was supplied at an essentially constant rate, while the other fluctuated. In theory, we have the alternative between a setting in which the biogenic carbonate component was brought in at a constant rate while the terrigenous clay component fluctuated, or one in which the terrigenous component remained essentially constant while the carbonate supply varied. The second possibility in turn poses two sub-alternatives: one in which the biogenic calcite production in the upper waters fluctuated; the other in which carbonate productivity remained essentially constant, while the chemistry of the bottom waters ranged from calcite-dissolving at some times to calcite-retaining at others.

The coincidence of the higher sedimentation rates with the more calcareous parts of the section, which also show the thickest limestone beds, suggests that the carbonate supply was the major variable, although we cannot yet assess the relative roles played by organic skeletal production and carbonate dissolution on the sea floor.

Profiles of trace metals (Fe, Ti, Ni, and Mn), silica, and alumina across bedding couplets show a maximum in shale interbeds, decreasing progressively toward centers of limestone beds. The total amount of iron in a given shaly interbed is approximately equal to that in the corresponding limestone. Assuming a constant rate of iron burial, we conclude that the time represented by the sedimentary couplet (20,000 to 100,000 yr) is divided about equally between the shaly phase and the calcareous one. Accordingly, the limestone beds accumulated at a mean rate more than twice that of the sequence as a whole, that is, at rates of more than 13 B, while the shales yield a very low apparent rate of sedimentation, less than 1 B. This figure does not take into account the diagenetic thinning of shales, either by compaction or by solution transfer of carbonates from the interbeds to the limestones, but a major contrast in depositional rates is indicated.

Bioturbation

The degree of bioturbation is influenced, among other factors, by oxidation state and rate of sedimentation. Oxygen availability limits the character and abundance of the bottom fauna, while sedimentation rate limits the time period for which the sediment is exposed to biogenic mixing. Both varied during the accumulation of the Gubbio section and, of the two, oxidation state has left the most pronounced imprint.

In the Scisti a Fucoidi, burrow-mottling of the lighter colored marlstones is common, but the black shales, rich in organic carbon, are usually laminated and unburrowed. Contrasts in colors and lithologies serve to render burrow mottling conspicuous here. Despite slow sedimentation in this interval, bioturbation did not destroy details of deposition (fine lamination, probably at the 10^3-yr level). Other gray to gray-green limestones and darker fissile interbeds within the Scaglia Bianca, and to some extent in the lower Scaglia Rossa, also were not thoroughly mixed by bioturbation, as evidenced by thin lamination and sparse burrow-mottling; such intervals remain relatively rich in organic carbon. Observations of depth of burrowing below significant marker beds, such as black shales having sharp contacts with underlying beds, suggest that burrowing after black shale deposition extended from 3 to 15 cm below the surface (usually less than 5 cm in these compacted marlstones).

Within the Campanian through Paleocene portion of the Scaglia Rossa, original laminations are virtually absent, and though some burrow-mottling is present, limestones and marls are basically homogeneous; bioturbation eliminated the details of depositional history. However, even here it was not effective enough to eliminate all of the history; the alternation of limestone and marly claystone survived. Near centers of lighter red-pink limestones, visible burrows are often filled with a darker red, more clay-rich sediment, evidence of primary sedimentary inhomogeneities which have not been completely eliminated.

Oxidation State

Oxidation state on the sea floor was primarily responsible for controlling the degree of bioturbation and for the diagenetic fixing of iron. Changes in oxidation state during deposition of the Gubbio sequence are reflected in variations in sediment coloration. These are influenced by retention of organic carbon and by the presence or absence of diagenetic pyrite or hematite. The state of iron is presumably dependent on original organic content and, to a smaller degree, on sedimentation rate.

Fischer and Arthur (1977) have shown that the transition from

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comparatively unoxidized Aptian-Albian sediments to more highly oxidized Cenomanian beds is of more than local significance and reflects world-wide changes in ocean dynamics.

LIMITATIONS ON MAGNETIC TIME RESOLUTION

The aptness of sedimentary sequences for paleomagnetic studies varies widely. Ignoring sediments whose magnetism is too weak to measure, and those which are magnetically unstable, the chief limitations of most sedimentary sequences lie in their lack of continuity. In shoal-water sediments or alluvium, by far the greater part of the time of deposition is represented by bedding planes, and a relatively small portion by the beds measurable in a sequence normal to stratification. Such primary interruptions do not seem to be a factor in the Gubbio sequence, which appears to have accumulated as a continuous rain of particles onto the sea floor — although this sedimentation was exceedingly slow during times of small carbonate supply.

The record of secular variation and polarity changes may have been essentially perfect in the initial sediment, but it was quickly modified by the stirring of burrowing animals. Most individual beds were thoroughly mixed by this process. While oceanic burrowing organisms can penetrate to a depth of a metre and more, thorough mixing in pelagic deposits is probably generally confined to the surficial 10 to 15 cm of uncompacted sediment (Guinasso and Schink, 1975), corresponding to perhaps 5 cm of consolidated rock. The retention of bedding at the decimetre level, at Gubbio, confirms this. The implication is that some details of the magnetic record for most of the Gubbio sequence were lost as a result of bioturbation, and the resolution was reduced to about 10,000 to 50,000 yr.

Subsequently, the magnetic record of the shaly interlayers was presumably further modified by compaction and by interstratal solution. Still later changes were probably superimposed on the physical and magnetic fabric of the shaly beds by shear, in the course of folding, and additional modification has probably occurred during weathering, in which the fractured shaly interbeds act as water conduits. As a result, the shaly interbeds, which now represent about one-half the time of each couplet, do not seem to be promising objects of magnetic inquiry — as the experience of Roggenthen and Napoleone (this issue) suggests.

The limestones, on the other hand, have preserved most of the physical fabric developed during the interval of bioturbation, which was essentially fixed by subsequent cementation. They therefore seem likely to retain much of the magnetic imprint which they carried at that time. Sedimentological considerations thus suggest that the maximal time resolution to be expected in this sequence of sediments is about one-half the duration of a bedding couplet, some 10,000 to 50,000 yr.

ACKNOWLEDGMENTS

The possible importance of the Gubbio section first came to Fischer's attention during a reconnaissance of pelagic sediments in the Tethyan realm, supported by a John Simon Guggenheim Memorial Fellowship. The work of the authors (Arthur, Fischer) was initiated with the help of a Princeton University Tuttle Bequest and was continued under the auspices of National Science Foundation Grant DES 74-22214.

Upper Cretaceous–Paleocene magnetic stratigraphy at Gubbio, Italy II. Biostratigraphy

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ABSTRACT

The upper Santonian to upper Paleocene pelagic sequence at Gubbio, Italy, yields a remarkably continuous succession of rich planktic foraminiferal faunas. Seven zones are recognized in sediments of the Late Cretaceous period: *Globotruncana concavata carinata, G. elevata, G. calcarata, G. tricarinata, G. gansseri, G. contusa,* and *Abathomphalus mayaroensis,* and the Paleocene is represented by "*Globigerina*" *eugubina, Globorotalia pseudobulloides, G. trinidadensis, G. uncinata, G. angulata, G. pusilla pusilla,* and *G. pseudomenardii.* These zones provide the biostratigraphic framework for the geomagnetic reversal sequence observed in the same section.

INTRODUCTION

This pelagic sequence is exposed along the Bottaccione Gorge near Gubbio. It ranges in age from Barremian to Eocene and contains a remarkably continuous succession of planktic foraminiferal faunas (Renz, 1936; Luterbacher and Premoli Silva, 1962). The purpose of the present contribution is to provide the basic biostratigraphic framework with which we have correlated the geomagnetic reversal sequence found in this section.

The first biostratigraphic subdivision of the Gubbio section was made by Renz (1936, 1951), who recognized six foraminiferal assemblage zones in the Aptian to Eocene interval. Later, Luterbacher and Premoli Silva (1962, 1964) and Luterbacher (1964) described and defined a detailed zonal scheme for the Tertiary portion of the section comparable to the tropical assemblages of that age. Cretaceous data went unpublished except for a report by Mohler (1966) who referred samples of Luterbacher and Premoli Silva and Renz to a compiled general zonation. Recently, some latest Cretaceous and Paleocene planktic foraminiferal zones have been reported from the Gubbio section in a preliminary attempt to relate magnetic stratigraphy to biostratigraphy (Premoli Silva and others, 1974).

As described in Arthur and Fischer (this issue), three main lithic units crop out in the Bottaccione Gorge (from bottom to top):

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the Scisti a Fucoidi, Scaglia Bianca, and Scaglia Rossa.

The present study has been conducted on two kinds of material: freed specimens from disaggregable (washable) samples and, more commonly, thin sections cut from indurated limestone or marlstone beds.

For the Cretaceous, thin sections were made at 1-m intervals. Within the Tertiary beds, they were taken at least than half-meter spacing corresponding to each bed sampled for paleomagnetic study. When present, disaggregable material was also collected along with underlying and overlying limestones. More than 800 samples were investigated.

Attempts to study calcareous nannoplankton (Mohler, 1966; B. Haq, 1974, personal commun.) were unsuccessful because of recrystallization and (or) dissolution of the coccoliths. Recently, Monechi and Pirini Radrizzani (1975) identified some forms in SEM scans of freshly fractured surfaces. The same authors are now studying the coccoliths, using samples collected during the present study.

BIOSTRATIGRAPHY

During Late Cretaceous to Paleocene time, the Umbrian region was located well within the tropical zone, and its rich faunal sequences are similar to those from other tropical oceanic areas at that time, such as the Caribbean and Trinidad. Therefore, faunal events recorded in the Gubbio section and in areas where a similar zonation has been established can be considered isochronous (Bolli, 1966; Pessagno, 1967).

In this paper, detailed zonation is limited to the late Santonian–Paleocene interval. A complete biostratigraphic study of the section is now in preparation by the author.

Fourteen zones, seven in the Paleocene and seven in the Upper Cretaceous, have been recognized in the Gubbio section. They comprise the Scaglia Rossa and the uppermost Scaglia Bianca. A detailed description of the faunal elements of these zones is given in papers by Premoli Silva and Bolli (1973) and Luterbacher (1964). Partial-range zones are most common, with a few total-range zones such as the *Globotruncana calcarata* Zone. Appearances and extinctions of the most important species providing the basis of the zonal scheme are shown in Figure 1. As a general rule, identification of species and zones can be made with greater accuracy from washed samples than from thin section. However, zonal markers, as well as most of the important

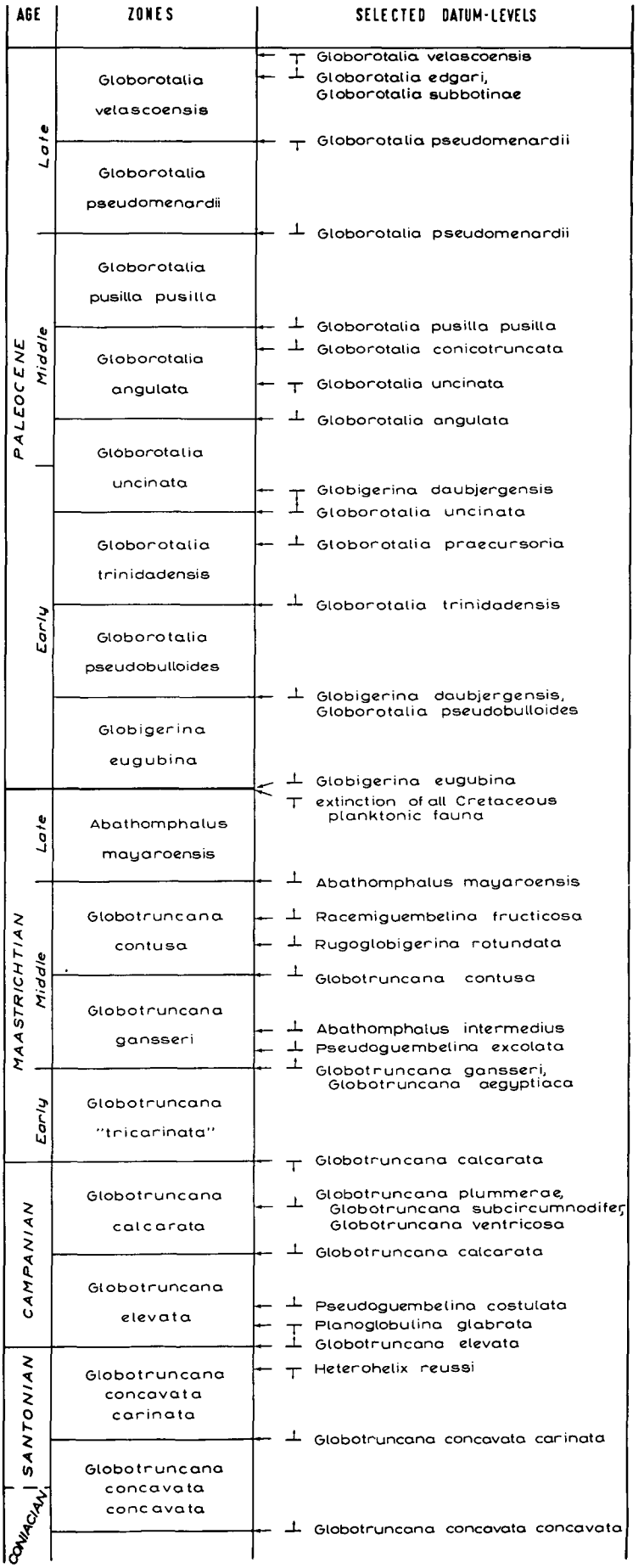


Figure 1. Late Cretaceous–Paleocene foraminiferal zonation and datum levels of index species: appearance, extinction (modified after Premoli Silva and Bolli, 1973).

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species in the Upper Cretaceous and Paleocene, have characteristic profiles which can easily be identified in two dimensions; the literature, particularly the European literature, contains photographs and drawings of oriented sections of most taxa (Postuma, 1971; Luterbacher, 1964; Pessagno, 1967). Use of several thin sections per sample, cut at various angles to the bedding, can provide a sufficient representation of the foraminiferal assemblage.

The zones identified are as follows (from top to bottom):

Paleocene

Globorotalia pseudomenardii Zone: from 371.5 to 364 m (7.5 m thick).
Globorotalia pusilla pusilla Zone: from 364 to 360.8 m (3.2 m thick).
Globorotalia angulata Zone: from 360.8 to 359 m (1.8 m thick).
Globorotalia uncinata Zone: from 359 to 356.5 m (2.5 m thick).
Globorotalia trinidadensis Zone: from 356.5 to 353 m (3.5 m thick).
Globorotalia pseudobulloides Zone: from 353 to 348 m (5 m thick).
Globigerina eugubina Zone: from 348 to 347.6 m (0.4 m thick).

Upper Cretaceous

Abathomphalus mayaroensis Zone: from 347.6 to 330 m (17.6 m thick).
Globotruncana contusa Zone: from 330 to 325.5 m (4.5 m thick).
Globotruncana gansseri Zone: from 325.5 to 302 m (23.5 m thick).
Globotruncana tricarinata Zone: from 302 to 270.2 m (31.8 m thick).
Globotruncana calcarata Zone: from 270.2 to 264 m (6.2 m thick).
Globotruncana elevata Zone: from 264 to 196 m, base not established.

CORRELATION BETWEEN BIOSTRATIGRAPHY AND MAGNETIC ANOMALIES

During the drilling program in the deep oceans (JOIDES–DSDP), sediments immediately overlying oceanic crust and related to marine magnetic anomalies have been recovered from a few sites. The few available dates have provided limited ties between paleontologic and magnetic scales; large uncertainties of correlation still exist, and revision and refinement are required.

The correlations which can be taken into account for the Late Cretaceous to Eocene time interval are reported in Table 1.

A composite paleomagnetic reversal sequence for the Maastrichtian, using data from DSDP sites 207A and 208 and from the Mendez Shale, Mexico, was recently suggested by Keating and others (1975) and correlated to the foraminiferal biostratigraphy. The main conclusions of their study were (a) anomaly 30 and the reversed interval just above it belong to the *Abathomphalus mayaroensis* Zone, (b) anomaly 31 is within the *Globotruncana gansseri* Zone (equivalent to *G. gansseri* and *G. contusa* of the present paper and), (c) anomaly 32b and anomaly 33b belong to the *Rugotruncana subcircumnodifer* Zone (equivalent to *G. tricarinata* of the present paper). Anomaly 33 is considered to mark the end of the Cretaceous Long Normal in earliest Campanian time.

TABLE 1. CORRELATIONS BETWEEN OCEAN-FLOOR ANOMALIES AND PALEONTOLOGICAL DATA

Anomalies	Nanno fossil zones	Foram zones	Sites	Age
24 (end)	*Marthasterites tribrachiatus*	*Globorotalia subbotinae*	39, center bit bottom of hole	Early Eocene
26 (begin)	*Discoaster mohleri*	*Globorotalia pseudomenardii*	213, core 17	Late Paleocene
30 (end)	*Micula mura?*	*Abathomphalus mayaroensis*	20C, core 6	Late Maastrichtian
31 (end)	*Micula mura*	*Not zoned*	239, core 19	Late Maastrichtian
33 to 34 Rev. Interval	*Eiffellithus eximius*	*Not zoned*	355, core 21	Early Campanian

Note: Data after Sclater and others (1974), Simpson, Schlich and others (1974), and Scientific Staff (1975).

CHRONOSTRATIGRAPHY

The absence of ammonites in the pelagic sequence at Gubbio and in general in the Late Cretaceous pelagic sediments in central and northern Italy prevents a direct correlation with the stages defined on the basis of this fossil group. Recent investiga-

AGE	AMMONITE ZONES	PLANKTONIC FORAMINIFERAL ZONES (present paper)
Maastrichtian	*Pachydiscus neubergicus*	*Abathomphalus mayaroensis*
	Acanthoscaphites tridens	*Globotruncana contusa*
		Globotruncana gansseri
		Globotruncana "tricarinata"
Campanian	*Bostrychoceras polyplocum*	*Globotruncana calcarata*
	Hoplitoplacenticeras vari	*Globotruncana elevata*
	Delawarella delawarensis	
	Placenticeras bidorsatum	
Santonian	*Placenticeras syrtale*	*Globotruncana concavata carinata*
	Texanites texanus	*Globotruncana concavata concavata*
Coniacian	*Parabevahites emscheri*	
	Barroisiceras haberfellneri	
Turonian Late	*Romaniceras deverlai*	*Globotruncana schneegansi*

Figure 2. Suggested correlation between the standard ammonite zonation and planktic foraminiferal zones (modified after Porthault, 1974; van Hinte, 1976).

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tions of the stratotypes and related sections of the Cretaceous stages provide biostratigraphic correlations among zonal schemes based on nannoplankton, foraminifera, and ammonites. Suggested correlation between ammonite and foraminiferal zonations based mainly on Porthault's data (1974) is shown in Figure 2, but more sophisticated correlations, using primarily the calcareous nannoplankton, are now available (Thierstein, 1977). In agreement with these more recent studies concerning the Upper Cretaceous, four zones can be attributed to the Maastrichtian (from top to bottom): *Abathomphalus mayaroensis, Globotruncana contusa, Globotruncana gansseri,* and *Globotruncana tricarinata;* two belong to the Campanian: *Globotruncana calcarata* and *Globotruncana elevata;* and the *Globotruncana concavata carinata* Zone belongs to the late Santonian.

Difficulties arise when correlations between foraminiferal zones and Paleocene stages are attempted. Although a number of papers on Paleocene stratotypes have been published since the early sixties, and more are pending, general agreement has not yet been reached. The Paleocene stratotypes are particularly unsuitable for providing complete sedimentary and fossiliferous sequences because of frequent, significant hiatuses. The writer prefers to not attempt even tentative correlations here. However, all seven zones used here for the Paleocene belong, by general agreement, to the Paleocene Epoch.

ACKNOWLEDGMENTS

The author is indebted to Lucia Paggi and Simonetta Monechi for their help in field work, to Ugo Antonucci for thin-section preparation, to M. B. Cita for her helpful comments, and to Anne Boersma for her assistance in revising the manuscript.

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Upper Cretaceous–Paleocene magnetic stratigraphy at Gubbio, Italy III. Upper Cretaceous magnetic stratigraphy

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ABSTRACT

Paleomagnetic study of the Upper Cretaceous part of the Scaglia Rossa pelagic limestone in the section at Gubbio, Italy, yielded a sequence of magnetic polarity zones that corresponds precisely with the polarity sequence inferred from marine magnetic anomaly profiles. This study shows the existence of a long normal polarity zone corresponding to the Cretaceous Quiet Zone, establishes the presence of the controversial interval of reversed polarity between marine magnetic anomalies 33 and 34, and places the Cretaceous-Tertiary boundary near the top of the reversed polarity interval immediately preceding anomaly 29.

INTRODUCTION

Paleomagnetic studies are proving to be an important tool for investigating the complex tectonic history of the Mediterranean region. The development of sophisticated, digital spinner magnetometers (Molyneux, 1972) and cryogenic magnetometers (Goree and Fuller, 1977) has permitted routine measurement of very weakly magnetic sedimentary rocks and thus greatly expanded the inventory of stratigraphic units that can be studied. This development led us, in 1973, to begin study of the Scaglia Rossa, a pink, pelagic, foram-coccolith limestone that represents the Late Cretaceous, Paleocene, and Eocene periods in the autochthonous or parautochhonous Umbrian Apennines of northern Italy. At about the same time, other groups of investigators began paleomagnetic studies of the Scaglia Rossa (Premoli Silva and others, 1974; Channell and Tarling, 1975; Klootwijk and Van den Berg, 1975).

Magnetic declinations at 20 localities in Umbria indicated an *average* counterclockwise rotation of the Italian Peninsula of about 45° (Lowrie and Alvarez, 1974). The most important section in the Umbrian Apennines is in the Bottaccione Gorge, near Gubbio (Arthur and Fischer, this issue). Detailed study of the variation in magnetic declination and inclination throughout the long interval represented by the Scaglia Rossa in the Gubbio section (about 50 m.y.: Cenomanian to middle Eocene) has shown that this part of Italy has probably undergone at least two counterclockwise rotations — 45° in the latest Cretaceous and a further 25° at some time after the middle Eocene (Alvarez and Lowrie, 1974; Lowrie and Alvarez, 1975).

Our measurements in the Gubbio section also revealed a sequence of normally and reversely magnetized zones, and in a series of successively more detailed sampling campaigns, we have been able to locate accurately the positions of the reversals in the Upper Cretaceous part of the section. In this paper, we give the combined results of all sampling through 1974, involving a total of 360 samples spaced at 0.5 to 1.0-m intervals. A further 140 samples were taken in 1975 in the vicinities of the reversal boundaries to locate each reversal as accurately as possible.

RESULTS

The magnetic mineralogy of the Scaglia Rossa limestone is believed to be dominated by magnetite, although hematite is also present at some sites. This conclusion was reached by Lowrie and Alvarez (1975) on the basis of Curie temperature analysis of extracted minerals and isothermal remanent magnetization acquisition curves on whole-rock samples. From thermal demagnetization studies, the effect of hematization on the original remanent direction was inferred to be minor at most sites, and this appears to be the case in the Cretaceous section at Gubbio.

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The mechanism of acquisition of the remanence of the pelagic limestone may be inferred by analogy with modern deep-sea sediments in which the most common magnetic mineral is ultra-fine–grained magnetite (Lovlie and others, 1971), and the remanence is acquired shortly after deposition by interstitial alignment of the magnetite grains with the ambient geomagnetic field (Kent, 1973). As a result, the remanent direction does not correspond exactly to the field at the time of deposition, but to a slightly later time when the remanence became blocked during consolidation (Løvlie, 1974). The time lag depends upon the sedimentation and consolidation rates, and in the Scaglia Rossa, probably represents about 10^4 yr. For this reason, the Scaglia Rossa probably records accurately the geomagnetic polarity inversions, but with an unknown, but geologically short, time lag in acquisition of remanence. In addition, the foraminiferal tests used for dating the limestone were mixed through bioturbation for some time after deposition, and so neither the remanent magnetization nor the final fossil assemblage remain unaltered from the time of deposition. Nevertheless, it seems probable that both attributes were firmly established by the time a given level had been buried a few tens of centimetres below the sediment-water interface. Thus the correlations between magnetic polarity and foraminiferal zones presented by Alvarez and others (this issue) are probably accurate to the nearest metre.

The natural remanent magnetizations (NRM) of most measured samples contained a large secondary component which completely masked the primary magnetization record. Progressive alternating field demagnetization in four successive fields was used to determine optimum directions, using the method of Symons and Stupavsky (1974). Magnetic cleaning in alternating fields of peak value 100 to 200 oe was generally sufficient to reduce the remanent

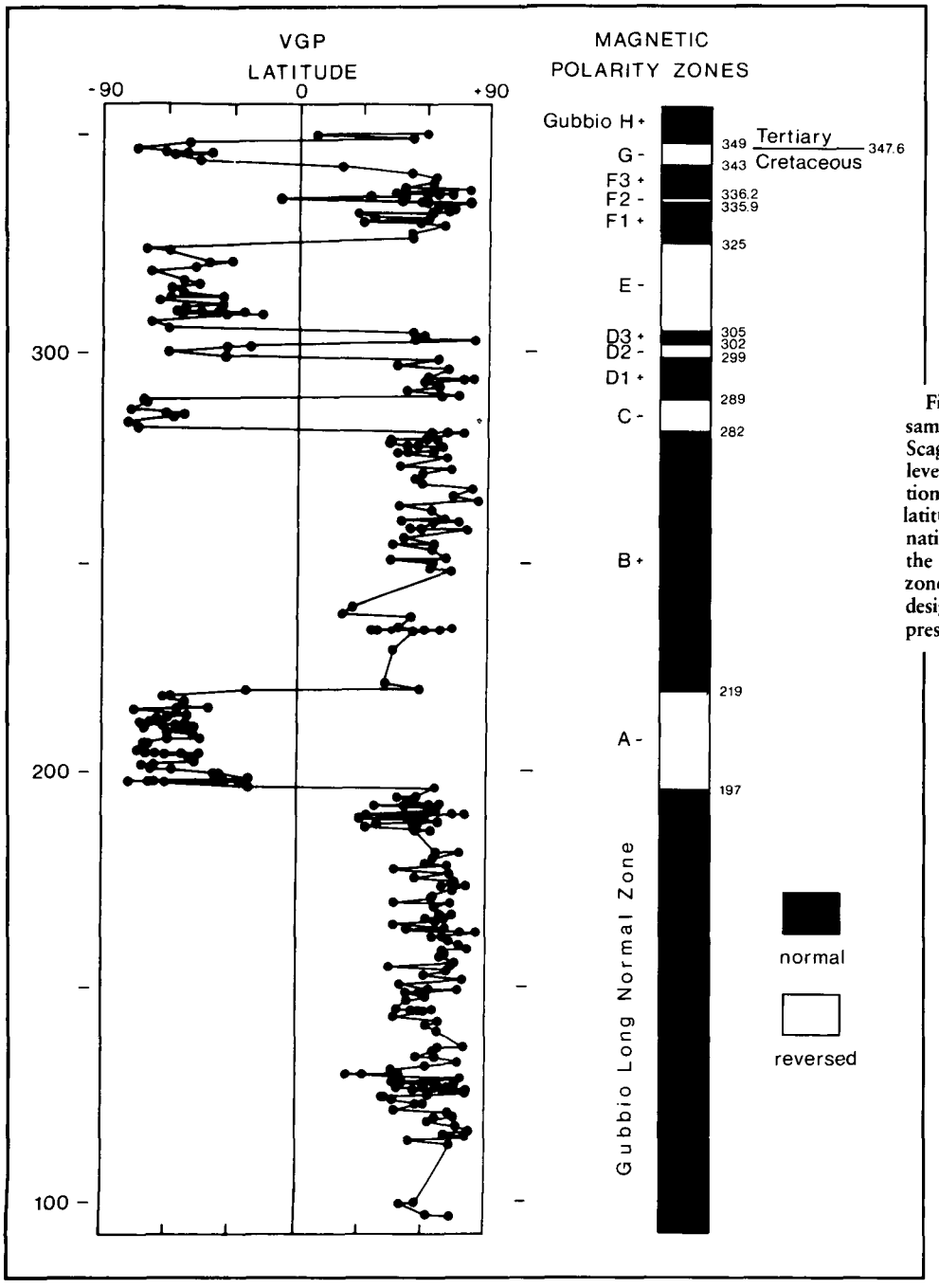


Figure 1. Magnetic results of metre-interval sampling of the Upper Cretaceous portion of the Scaglia Rossa section at Gubbio. Stratigraphic levels in metres correspond to the measured section of Arthur and Fischer (this issue). VGP latitude for each sample is computed from inclination and adjusted declination as explained in the text. At the right is the sequence of polarity zones based on these data. The polarity zones are designated according to the alphabetical system presented by Alvarez and others (this issue).

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intensities by an order of magnitude (to an average of 3×10^{-7} G) and to isolate the characteristic remanent directions.

The mean direction of stable remanent magnetization from the Upper Cretaceous part of the Gubbio section is: declination = 304°, inclination = 43.9°, α_{95} = 2.6°. The inclination is almost identical with the value of 43.7° predicted for the sampling site in the Late Cretaceous from the paleoreconstruction of Van der Voo and French (1974). Because of the absence of Tertiary samples in the present data set, the mean declination is more westerly than the value obtained by Lowrie and Alvarez (1974) from a paleomagnetic study of the Scaglia Rossa throughout Umbria. The declinations show the effects of the large-scale counterclockwise tectonic rotations experienced by the Northern Apennines during and after deposition of the Scaglia Rossa.

The data were adjusted to compensate (a) for the known tectonic rotation and (b) for the difference in latitude between the site at present and at the time of deposition. First, the declinations were rotated clockwise by 54° so as to set the mean declination equal to that expected from the Van der Voo and French (1974) paleoreconstruction (358°). Then the paleolatitude of the site was determined from the mean inclination. This, rather than the present site latitude, was used to compute a virtual geomagnetic pole (VGP) position for each sample on the basis of its measured inclination and corrected declination.

The VGP latitude indicates the geomagnetic polarity during deposition; its variation for all optimally demagnetized samples from the Upper Cretaceous part of the Gubbio section is shown in Figure 1. The stratigraphic position of the samples follows the measured section of Arthur and Fischer (this issue) to facilitate comparison with the magnetic measurements of Roggenthen and Napoleone (this issue) and the paleontological zonation determined by Premoli Silva (this issue). The resulting geomagnetic polarity column is shown at the right in Figure 1. The polarity zones are designated alphabetically, using the system adopted by Alvarez and others (this issue).

DISCUSSION

In Figure 2, the sequence of polarity intervals based on the section at Gubbio is compared with marine magnetic anomaly profiles and their interpreted reversal sequences from three oceanic areas where the boundary between the Cretaceous Quiet Zone and the

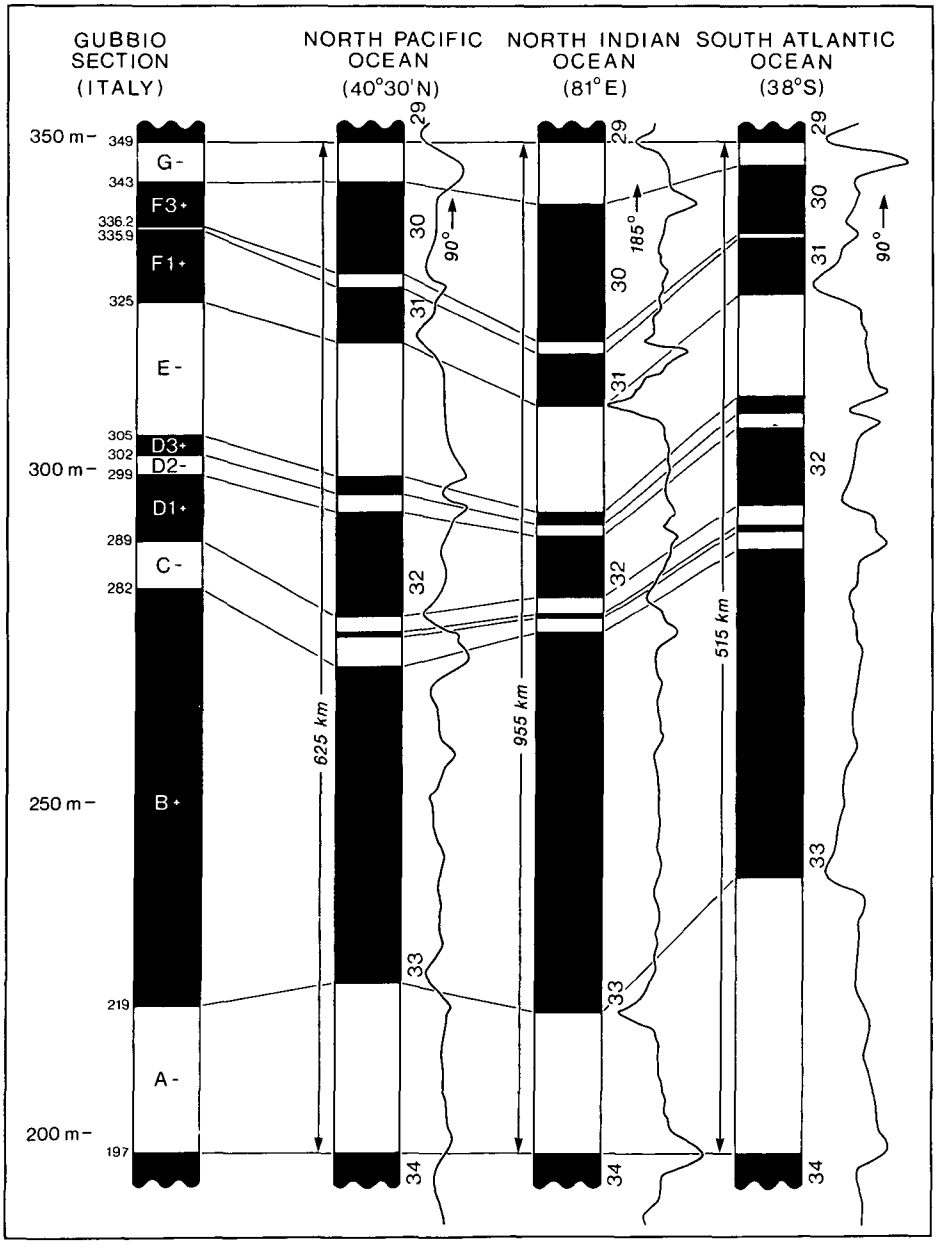


Figure 2. Comparison of the Late Cretaceous geomagnetic polarity sequence of the Scaglia Rossa section at Gubbio (Fig. 1) with marine magnetic profiles and interpreted geomagnetic polarity sequences from three oceanic areas where the reversed interval between anomalies 33 and 34 can be recognized. The oceanic profiles have been uniformly reduced to bring the reversal boundaries at the beginning and end of the sequence into coincidence with the same horizons in the Gubbio profile. North Pacific Ocean: profile B of Raff (1966); North Indian Ocean: profile V1909 (= V19 of Sclater and Fischer, 1974, Fig. 2); South Atlantic Ocean: profile C1102 (1) of Ladd (1974, p. 101), extended to anomaly 34 by Lamont-Doherty profile V3101. Reversal boundaries on all three profiles were picked by R. L. Larson (1975, personal commun.). Note that the oceanic columns tie sideways to their respective anomaly profiles, not diagonally along the correlation lines.

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Late Cretaceous–Cenozoic anomaly sequence can be recognized. The reversal sequences based on marine magnetic measurements have been reduced uniformly in order to align the "base" of anomaly 29 and the "top" of the Cretaceous Quiet Zone with the analogous horizons in the Gubbio column.

The match between the Gubbio column and the three marine reversal sequences is unmistakable and establishes beyond question that the pelagic sediments of the Scaglia Rossa and the oceanic crust in these three areas were recording the same sequence of geomagnetic field reversals.

Although the pattern of long and short polarity intervals is clearly the same in all four columns, none of the columns exactly matches any other. This is to be expected, because of the probability of independent variations in sedimentation rate in the Gubbio section and in spreading rate in the three oceanic areas. Comparison of the three oceanic profiles suggests at least two or three changes in spreading rate in each ocean over the approximately 20-m.y. time span. It is clear from Figure 2, however, that there have been no lasting order-of-magnitude differences in either sedimentation rate at Gubbio or in spreading rate in the oceanic areas examined.

Although the lowest portion of the Paleocene is missing in most of the known stratigraphic sections in the world, Luterbacher and Premoli Silva (1964) have shown that the Gubbio section is essentially continuous across the Cretaceous-Tertiary boundary. This boundary can be picked very precisely (even with a hand lens) by the disappearance of large globotruncanids, and it falls at 347.6 m in the measured section of Arthur and Fischer (this issue). This is almost at the top of the reversed polarity zone Gubbio G−[1], which corresponds, in the marine magnetic profiles, to the reversed segment immediately preceding anomaly 29. This is in agreement with Sclater and others (1974), who placed the Cretaceous-Tertiary boundary "between 29 and 30."

The data from the lower part of the Upper Cretaceous section indicate constant normal polarity of the geomagnetic field. They provide confirmation that the Cretaceous Magnetic Quiet Zone as seen on marine magnetic anomaly profiles is indeed due to the absence of polarity inversions during this long interval and that this quiet zone, at least, is not attributable to breakdown of the recording mechanism in the oceanic crust.

The interpretations of the marine profiles (Fig. 2) show a very short normal interval within the reversed interval immediately preceding anomaly 32, but we have not yet observed this short normal in the Gubbio section (Fig. 1). It is possible that it occurs between two samples, although sample spacing is no greater than 1.5 m in this part of the section. Measurement of recently taken, closely spaced samples should establish whether this zone is present or not. Very small faults were noted near the middle of reversed zone Gubbio C−, but these are probably too minor to account for the absence of the short normal. An alternative explanation is that the minor deflection of the marine magnetic profiles within the reversed interval represents a field excursion or a short-lived decrease in field intensity rather than a full reversal. The variation of VGP latitude within the Gubbio C− Zone (Fig. 1) is suggestive of an excursion, but it may just as well be due to statistical fluctuation, as seen in the Gubbio A− Zone.[2]

An important aspect of our results concerns the end of the Cretaceous Long Normal Interval (corresponding to the Cretaceous Quiet Zone observed in marine magnetic studies). The first published time scale of Late Cretaceous–Cenozoic magnetic reversals (Heirtzler and others, 1968) did not extend back farther than anomaly 32 (Fig. 2), but Helsley and Steiner (1969) suggested the existence of a long normal interval in the Cretaceous, prior to anomaly 32. Larson and Pitman (1972, Fig. 5) showed two additional intervals of reversed polarity between the base of Heirtzler and others' (1968) time scale and the top of the Cretaceous Long Normal, and they designated a new anomaly (33) just above the lowest reversed interval. (Although this was not discussed in the paper, the two new reversed intervals were based on observations in the northeast Pacific, the southwest Atlantic, and the northern Indian Ocean; R. L. Larson, 1975, personal commun.) In subsequent informal usage, anomaly 34 was added at the top of the Cretaceous Long Normal, with anomalies 33 and 34 bracketing the first reversed interval above the Long Normal. In a map of world-wide marine magnetic lineations (Pitman and others, 1974), anomaly 34 was shown in the South Atlantic Ocean. These new reversed intervals were not universally accepted; for example, Sclater and Fischer (1974) showed the polarity intervals back to reversed interval 33b (= Gubbio C −), but cut their time scale off without showing the reversed interval of anomalies 33-34 (= Gubbio A−).

Doubts over the existence of the reversed interval between anomalies 33 and 34 were increased by Shive and Frerichs' (1974) study of the paleomagnetism of the Niobrara Formation in the western United States, for which related foraminiferal studies (Frerichs and Adams, 1973; Adams and Frerichs, 1973) indicated a Coniacian to early Maastrichtian age. About 350 samples distributed over four sections through the Niobrara failed to yield a single reversed direction, although the dating suggested that at least part of the Late Cretaceous reversal sequence should be contemporaneous with the Niobrara. The possible explanations considered by Shive and Frerichs were (1) misdating of the marine magnetic anomalies, (2) misdating of the Niobrara Formation, (3) hiatuses in sedimentation or sampling gaps in the Niobrara, (4) remagnetization of the Niobrara during a time of normal polarity, and (5) the possibility that some of the Late Cretaceous anomalies might not represent full field reversals. Nevertheless, they found arguments against all five of these possible explanations, and therefore, when Keating and others (1975) compiled a Late Cretaceous reversal sequence based on paleomagnetic studies of sedimentary sections, they incorporated the Niobrara data and did not show the reversed interval between anomalies 33 and 34.

Frerichs and Adams (1973, p. 190) noted that they were assigning a much younger age to the Niobrara Formation than was previously accepted, but deferred discussion of the age revision to a future paper. If, however, one accepts the standard dating of the Niobrara as Coniacian-Santonian (Cobban and Reeside, 1952), the unbroken normal polarity is to be expected, since, on the basis of the Gubbio results (Alvarez and others, this issue), the Niobrara would have been deposited entirely during the Cretaceous Long Normal Interval.

This may or may not be the reason for the unbroken normal polarity in the Niobrara Formation, but in view of the excellent match between the Gubbio profile and the marine magnetic anomalies, it is difficult to entertain further doubts over the existence of an interval of reversed polarity bounded by anomalies 33 and 34.

ACKNOWLEDGMENTS

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[1]Polarity zones are designated by letters with the prefix "Gubbio" in ascending alphabetical order. Normal zones are distinguished by "+" and reversed zones by "−" (see Alvarez and others, this issue).

[2] Subsequent sampling has shown that a short normal interval is in fact present within the Gubbio C− interval.

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Upper Cretaceous–Paleocene magnetic stratigraphy at Gubbio, Italy IV. Upper Maastrichtian-Paleocene Magnetic Stratigraphy

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ABSTRACT

Pelagic limestones ranging in age from mid-Maastrichtian through Paleocene yield several well-defined magnetic polarity zones. These zones are tied to the planktic foraminiferal zonation at this site and are correlated with magnetic anomalies 26 through 31.

INTRODUCTION

The general nature of the stratigraphic sequence at Gubbio has been described by Arthur and Fisher; its biostratigraphy by Premoli Silva, and the magnetic stratigraphy of the Upper Cretaceous by Lowrie and Alvarez (this issue). Our study was restricted to the interval between metres 321 and 368 of the section described by Arthur and Fischer (paper I), overlapping the work of Lowrie and Alvarez by 26 metres; we present the magnetic data obtained from the upper Maastrichtian and the Paleocene. General conclusions are confined to the following paper (V) by all of the authors.

THE SEDIMENTARY SEQUENCE

The section studied extends from the middle Maastrichtian (*gansseri* zone) to near the top of the Paleocene (*pseudomenardii* zone). Except for tilting (present attitude is about N50°W dipping 35° to 55°NE), the rocks and fossils show no evidence of pervasive deformation. Complications are confined to a few small-scale intrastratal folds and some minor penecontemporaneous slumping.

The sequence studied consists of pink to red pelagic coccolith-foraminiferal limestones in medium to thin beds (<20 cm), separated by thin marly interbeds (<10 cm). The limestones show little to no internal layering, but burrow mottling is common. The marly interbeds seem to have resulted from a combination of factors: (1) a primary variation in clay content, and (2) a concentration of secondary intrastratal solution along the more clay-rich layers.

Sampling was almost entirely confined to the limestone beds, since only these can be readily cored in outcrop. Toward the top of the Paleocene, however, where the limestone beds are very thin and marly layers and stylolites are closely spaced, samples are commonly less homogeneous and contain some of this more clay-rich material.

Although the Scaglia Rossa is remarkably

TABLE 1. PALEOCENE AND MAASTRICHTIAN DIRECTIONS FOR THERMAL AND AF 150 oe DEMAGNETIZATION

	N	Dec.	Inc.	k	a_{95}
Paleocene thermal	93	326.	+42.	7.1	5.9
Maastrichtian thermal	35	308.	+43.	9.9	8.1
Paleocene AF 150 oe	141	332.	+25.	2.8	9.0
Maastrichtian AF 150 oe	129	327.	+45.	2.3	10.8
Gubbio L+	17	327.	+31.	8.0	13.5
Gubbio K−	12	150.	−48.	3.9	25.3
Gubbio M−	35	160.	−44.	10.6	7.8

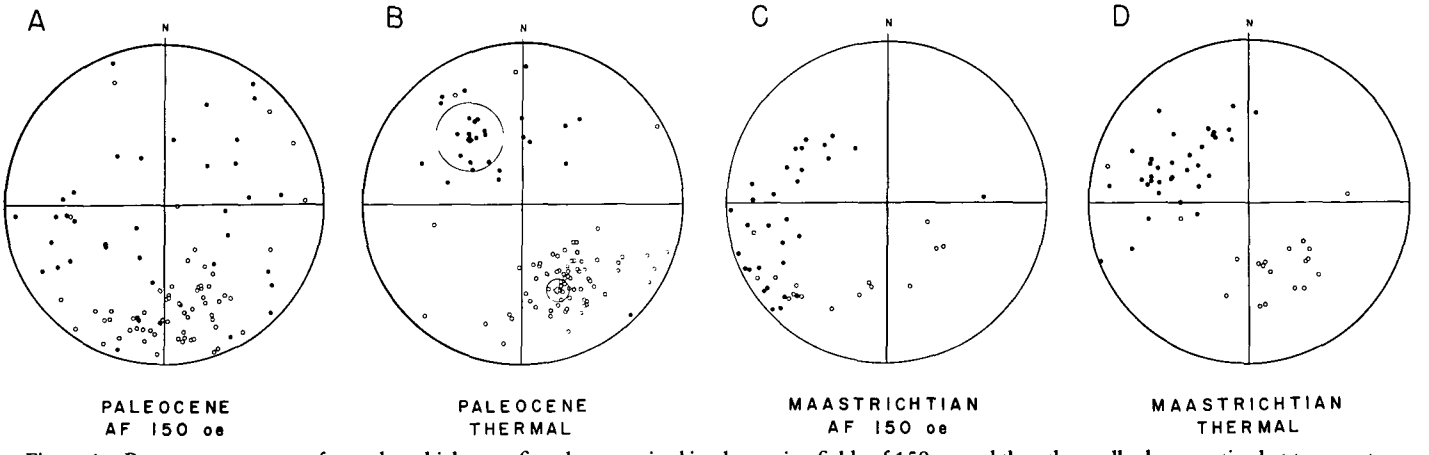


Figure 1. Remanence vectors of samples which were first demagnetized in alternating fields of 150 oe and then thermally demagnetized at temperatures of 350° ± 50°C. A. All Paleocene samples used in the delineation of the polarity zones after being AF demagnetized to 150 oe. B. The same Paleocene samples after being thermally demagnetized. Directional averages and α_{95} cones of confidence are also shown. C. AF demagnetization of only those Maastrichtian samples which came from portions of the section showing intermediate directions. D. Thermal demagnetization of the Maastrichtian samples.

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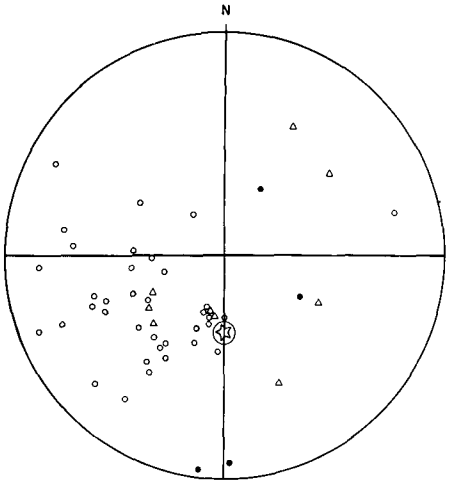


Figure 2. Vector subtraction of thermal directions from AF directions. Open symbols (○, Δ) indicate negative directions; filled symbols indicate positive directions (●,▶). Open triangles (Δ) indicate subtracted vectors of those samples judged to be of "reverse" polarity from Figure 1D; and the open circles (○) indicate those vectors derived from samples of "normal" polarity. The directions are prior to a correction for bedding. The star represents the present-day field if it were reversed.

homogeneous throughout its vertical extent, the apparent sedimentation rate slowed from the Maastrichtian to the Paleocene. The average apparent sedimentation rate dropped from 13–15 m/my. to 2.5–3.0 m/m.y. of compacted rock (Arthur and Fischer, this issue, Table 1). The sedimentation rates calculated are "apparent" because accurate means of establishing effects of compaction and dissolution are lacking. The differences in sedimentation rate manifest themselves in thinner limestone beds and in a correspondingly greater frequency and proportion of marly beds in the Paleocene.

SAMPLING AND ANALYTICAL PROCEDURES

Except for a small pilot group collected as block samples, 1-in. cores were collected in the field with a water-cooled drill. Cores were oriented by magnetic compass and, in

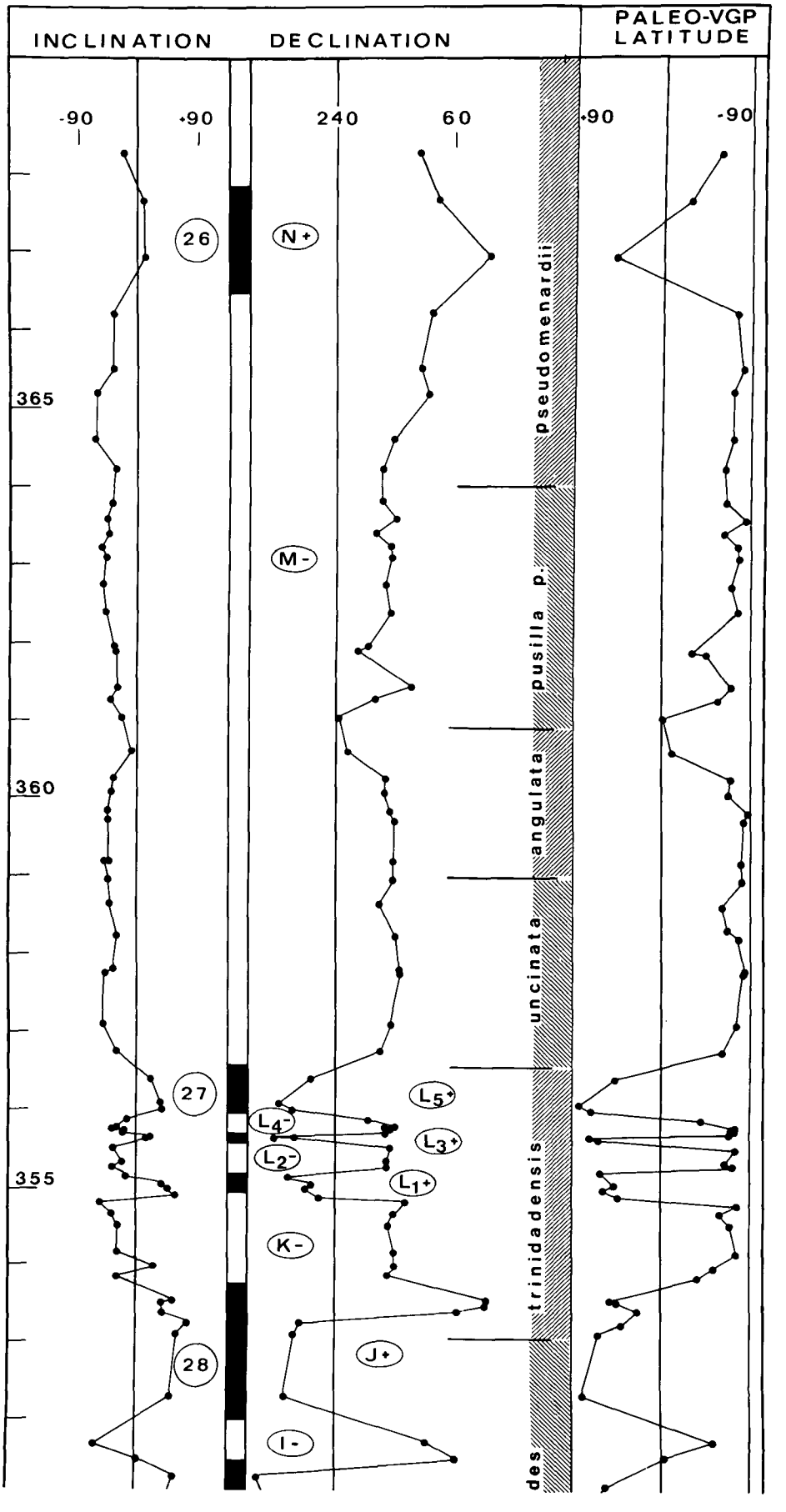


Figure 3. Stratigraphic distribution of the polarity zones relative to the measured section of Arthur and Fischer (this issue) and to the planktic foraminiferal zonation of Premoli Silva (this issue). Declinations representing reversed and normal polarities are separated by the 240°-60° azimuth (rather than a 270°-90° azimuth), which reflects the rotation of the Italian Peninsula. Filled circles (●) are thermally demagnetized samples; filled triangles (▶) have been only AF treated. The paleo–VGP latitudes are calculated with respect to the mean paleomagnetic pole of the entire section investigated (see text).

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some instances where the roadcuts are protected with wire netting, by sun compass. Sample groups were collected at approximately 1-m intervals in the Maastrichtian and .1- to .5-m intervals close to the Tertiary-Cretaceous boundary and in the Paleocene sediments. The 47-m section yielded 375 samples from a total of 148 separate beds

Measurements of the magnetic directions were made using 1-in.-long rock cylinders. One-third of the analyses following AF demagnetization were made with a cryogenic magnetometer at the University of Pittsburgh, and the remainder of analyses, including those made after thermal demagnetization, were made on a cryogenic magnetometer at the University of Texas at Dallas inside a mu-metal shielded room.

DEMAGNETIZATION AND SEDIMENT MAGNETIZATION

The presence of magnetite in the Scaglia Rossa sediments was demonstrated by Lowrie and Alvarez (1975) and Van den Berg and others (1975). Additional evidence was noted by Mauritsch and Turner (1975), who observed a drop in IRM when a sample was allowed to heat through the low-temperature transition in the magnetocrystalline anistropy constant (K_1) which occurred at −143°C (130° K). While hematite is also present in most of these rocks, we conclude that the remanent magnetism observed resides primarily in very fine grained magnetite similar to that of some chalks and oozes on the deep sea floor (Løvlie, 1974; and Opdyke and Foster, 1970).

The natural remanent magnetization vectors are very scattered, but directional grouping improves upon application of AF demagnetization at peak fields of 150 oe, as was found for these rocks by Premoli Silva and others (1974) and for the Scaglia Rossa in general by Lowrie and Alvarez (1975). Some of the Maastrichtian and most of the Paleocene samples, however, are still not satisfactorily grouped after the AF treatment (Figs. 1A, 1C). Thermal demagnetization of these same samples at moderate temperatures (350° ± 50°C), moved most of the scattered directions into agreement with the bulk of the samples to give better grouping, as shown in Figures 1B and 1D.

All of the Paleocene samples used in the following determination of polarity zones and the portions of the Maastrichtian showing intermediate directions after AF treatment (Fig. 1C) were thermally demagnetized. The vectors removed by the thermal treatment are plotted in Figure 2, prior to the correction for bedding. They are remarkably consistent, considering that these directions are found by subtracting two directions whose intensities are already very

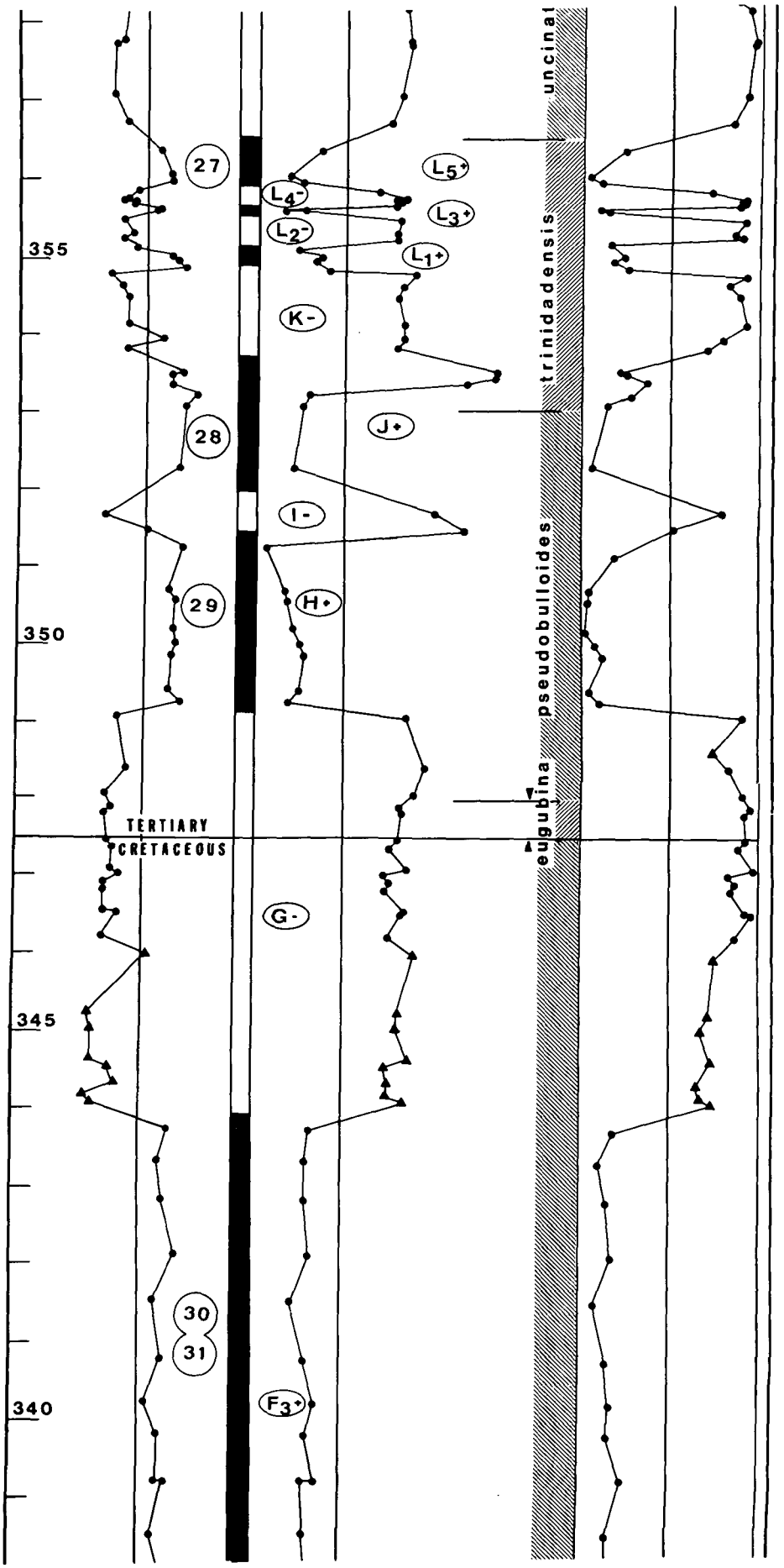
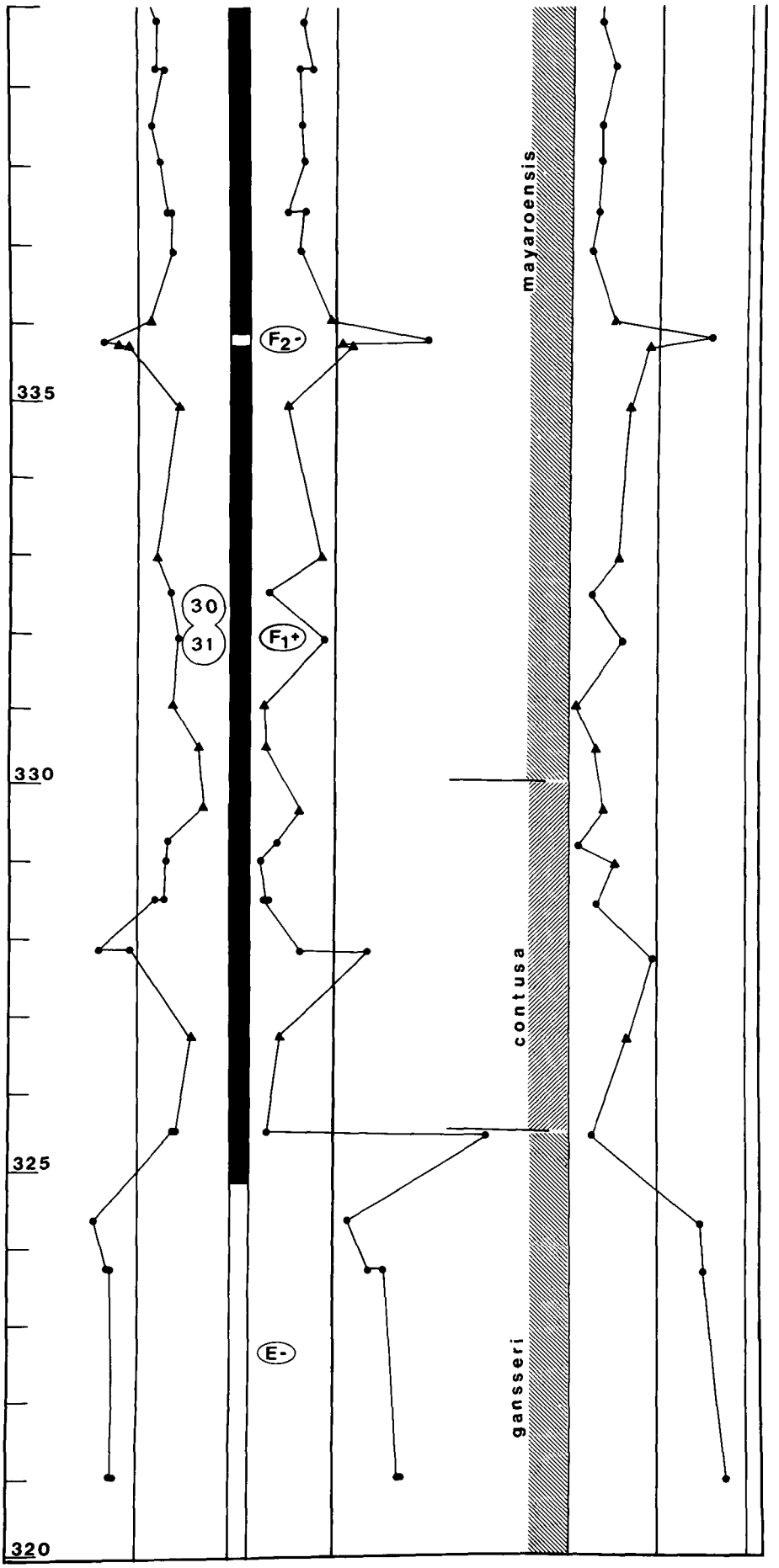


Figure 3. (*Continued*).

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small (in the range of 10^{-5}emu/cm^3) and that some of the original, primary magnetization is probably being removed along with the overprint. Almost all of the subtraction vectors (AF at 150 oe minus thermal demagnetization directions) are negative, and many group in the southwest with a relatively high inclination. The star represents the position of the present-day field if it were reversed. Although not conclusive, this suggests that the overprint which is resistant to AF treatment was acquired after the beds were tilted to their present position and is predominantly the result of a reversed field. The average directions for the Paleocene and Maastrichtian are listed in Table 1 along with selected portions of the Paleocene which will be discussed later. In view of the fact that sample spacings were relatively even, all directions in Table 1 have been simply averaged together. Gubbio zones L+, J−, and M− are based on thermally demagnetized samples. Zones L+ and J− are from more marly parts of the Paleocene, and M− is from more carbonate-rich beds.

Bioturbation has destroyed the original depositional fabric and any primary detrital remanence as well, with the consequent disruption of very short events. Therefore the measured remanence was acquired after the cessation of the burrowing activity. Arthur and Fischer (this issue) estimated that in the limestones, burrowing stirred the sediment thoroughly to a depth of at least .05 to .1 m (.1 to .2 m in the unconsolidated sediments). The limestone beds, in turn, represent only part of the depositional history. They therefore may reflect only segments of the magnetic history as well. This is borne out by the abrupt changes in magnetic directions found between successive beds only .1 to .05 m apart. Although these polarity changes may conceivably be the result of differential acquisition of magnetization, and the remanence directions measured were acquired prior to tectonic rotation (Table 1). Zones L+ and J− are in two of the more marly portions of the section, and their average directions compare favorably with those of M−, which is more carbonate rich, and which corresponds to the average direction of the Paleocene.

RESULTS

Six normal and six reversed zones are defined (Fig. 3). One of the zones appears to be interrupted by two small subzones.

The vectors derived from thermal demagnetization are plotted in Figures 1 and 3 along with vector averages. The average directions correspond closely to the magnetic directions obtained by Lowrie and Alvarez (1975) for Scaglia Rossa samples at Gubbio and other parts of Umbria, demon-

Figure 3. (*Continued*).

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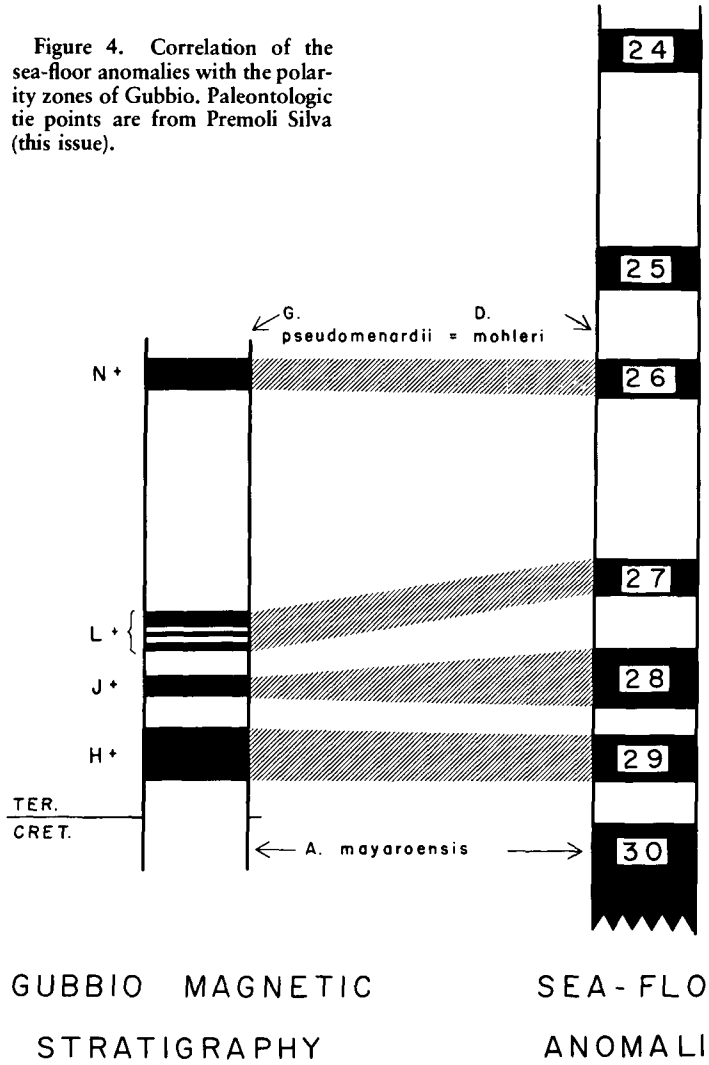


Figure 4. Correlation of the sea-floor anomalies with the polarity zones of Gubbio. Paleontologic tie points are from Premoli Silva (this issue).

strating a counterclockwise rotation of the northern Italian peninsula (Table 1).

The stratigraphic distribution of polarity zones relative to the measured section and to the biostratigraphic zones (Premoli Silva, this issue) is shown in Figure 3. Because of the tectonic rotation, the reference line chosen as evenly separating normal and reversed magnetic values is the 60°-240° azimuth. To the right of the declinations are plotted the paleo–Virtual Geomagnetic Pole (paleo–VGP) latitudes for each bed sampled.

Polarity changes through a section are perhaps most simply shown by plots of VGP latitudes (Watkins and Baksi, 1974). In calculating the paleo–VGP latitudes, a mean paleomagnetic pole was determined for the section studied (321 to 368 m). The angular difference between this pole and the VGP for any one bed yields the paleocolatitude for that bed. The plotted latitudes, then, reflect both the declinations and the inclinations and are independent of subsequent structural rotations.

DISCUSSION

The six normal and six reversed magnetic polarity zones recognized in the section have been named in accordance with the system explained in Alvarez and others (paper V, this issue) and range from Gubbio F+ to Gubbio N+. Two of them, Gubbio F+ and Gubbio L+, are subdivided into subzones by minor reversals (Fig. 3).

The reliability of the magnetic directions throughout the section is confirmed by the observation that the reversed and normal directions are essentially antiparallel and that they consistently show a similar counterclockwise rotation after bedding correction. In the part of the section that has been lumped in the zone Gubbio L+, the magnetic record shows much fluctuation. This particular part of the section is exceptionally rich in clay (although still nominally a limestone), and it probably represents a condensed section.

Toward the top of the Paleocene, the limestone layers become much thinner, and the redder, more clay-rich beds become more frequent. Many of the samples from this part of the section have magnetic directions close to the present field and are little affected by AF fields up to 700 oe or by thermal demagnetization up to 600°C. Therefore, the study was terminated in the *pseudomenardii* zone and not carried to the very top of the Paleocene.

In general, magnetic reversals do not coincide with either lithic or faunal boundaries. An exception to this occurs just below 357 m. Here a prominent change in lithology (from a marly, fissile limestone below to a more massive limestone above) coincides with the polarity boundary between Gubbio L+ and Gubbio M−. This also corresponds to the base of the uncinata zone. This coincidence may have been brought about by a minor break in sedimentation.

Correlation of our magnetic zones with the sea-floor anomalies (Fig. 4) was accomplished by matching both the anomaly pattern and two paleontologically dated anomalies: (1) anomaly 26 seems to be coincident with the *Globorotalia pseudomenardii* zone which corresponds to the *Discoaster mohleri* zone (Premoli Silva, this issue); (2) the Tertiary-Cretaceous boundary seems to lie just below anomaly 29 (Sclater and others, 1974). The long reversed zone Gubbio M− must correspond to a long interval in the sea-floor pattern — either that below anomaly 26 or that below anomaly 24. These constraints yielded only one solution, that shown in Figures 3 and 4. The small episodes below Gubbio M− were lumped together on the assumption that they probably would not have been recognizable from the sea-floor pattern because of their short durations. The correlation scheme is summarized in Figure 4. This correlation was further substantiated when it was found to agree precisely, in the areas of overlap, with the independent correlations of Lowrie and Alvarez (this issue) who began matching from the Gubbio Long Normal Zone farther down in the section.

ACKNOWLEDGMENTS

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Upper Cretaceous–Paleocene magnetic stratigraphy at Gubbio, Italy V. Type section for the Late Cretaceous–Paleocene geomagnetic reversal time scale

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ABSTRACT

A biostratigraphically complete and well-exposed sequence of Upper Cretaceous–Paleocene pelagic limestones at Gubbio, Italy, has provided a record of geomagnetic polarity reversals that closely matches the sequence inferred from marine magnetic anomalies. Abundant foraminifera permit accurate dating of the sequence. Because of these favorable circumstances, the Gubbio locality is formally proposed as the magnetostratigraphic type section for the Upper Cretaceous and Paleocene.

INTRODUCTION

The realization that the Earth's magnetic field has undergone repeated reversals in polarity was in itself a major advance in geophysics. In addition, as the effects of geomagnetic reversals are globally synchronous, they provide geologists with a powerful new correlation tool for the study of Earth history. Geomagnetic polarity is imprinted on various rocks in the form of remanent magnetism, and the history of the geomagnetic field may be deciphered from two independent records: (1) marine magnetic anomalies resulting from magnetic imprints on laterally accreted oceanic crust and (2) the magnetic stratigraphy of vertically accreted sequences of sedimentary and volcanic rocks. In order for magnetic stratigraphy to reach its full potential, these two records need to be precisely correlated with each other and with the biostratigraphic time scale, and to be calibrated in terms of absolute age by correlation with radiometrically dated rocks.

In the four preceding papers, we have given the results of lithostratigraphic, paleontologic, and paleomagnetic studies of what appears to be a continuously deposited sequence of Upper Cretaceous and Paleocene sediments at Gubbio, Italy. In this paper, we tie these three kinds of investigation together in order to obtain accurate paleontological dating for the reversal sequence at Gubbio and, by correlation, for the marine magnetic anomaly pattern. Recognizing the need for a firm stratigraphic basis for inferred magnetic history, we propose the Gubbio section as a magnetostratigraphic type section for the Upper Cretaceous and Paleocene.

DATING SEA-FLOOR ANOMALIES

A number of pioneers (Cox and others, 1963a, 1963b; McDougall and Tarling, 1964; Doell and Dalrymple, 1966; McDougall and Chamalaun, 1966) working on late Neogene volcanic sequences, established in the 1960s the reality of correlatable *magnetic polarity zones*, and inferred that these had resulted from periodic reversals of the geomagnetic dipole. Vine and Matthews (1963) recognized that the anomaly stripes on magnetic maps of ocean floors resulted from lateral accretion of oceanic crust in this reversing geomagnetic field, and that the sea floor potentially reveals the history of these reversals. On this basis, Heirtzler and others (1968) proposed the first comprehensive polarity chronology extending back to the Late Cretaceous. Helsley and Steiner (1969), Larson and Pitman (1972), Larson (1974), and a number of other workers have delineated four major episodes in geomagnetic history since the breakup of Pangaea in the early Mesozoic (Table 1). These major episodes, now established in various ocean basins, have led to refinement of paleogeography and to partial understanding of the spreading history of oceanic crust.

A precise history of the oceans requires dating of the crustal strips which generate the individual anomalies, and this has not yet been achieved to the desired accuracy. Direct methods useful in dating these crustal strips are: (1) determining the radiometric age of basaltic basement rocks recovered beneath known anomalies in DSDP holes, (2) dating the oldest overlying sediments by paleontological means, and (3) interpolating between and extrapolating beyond well-dated anomalies, assuming a constant spreading rate. Each of these approaches has limitations. Samples are few, and many of them are not suitable for radiometric dating. The oceanic basement is highly complex, with superimposed extrusive lavas and intrusive dikes and sills whose ages may differ from the country rock. Furthermore, alteration is common. Consequently, single radiometric dates may be misleading. Paleontologic dates are too young if the oldest fossiliferous beds were preceded by unfossiliferous deposits or are separated

TABLE 1. MAJOR EPISODES IN GEOMAGNETIC FIELD BEHAVIOR

Episode	Marine magnetic anomaly pattern	Geomagnetic field behavior
4	Cretaceous-Cenozoic anomaly sequence	Cretaceous-Cenozoic reversal sequence (Late Cretaceous to present)
3	Cretaceous Quiet Zone	Cretaceous Long Normal polarity chron (Early to Late Cretaceous)
2	Keathley, Japanese, Hawaiian, and Phoenix magnetic lineations	M-sequence reversals (Late Jurassic to Early Cretaceous)
1	Jurassic Quiet Zone	Field behavior uncertain

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from the underlying basement by a depositional hiatus, or if the hole bottoms in a sill instead of in true basement. Finally, variations in spreading rates have been demonstrated over time intervals on the order of 10^7 yr (Sclater and others, 1974; Larson and Pitman, 1972; Hays and Pitman, 1973; Lowrie and Alvarez, this issue, Fig. 2). The fundamental problem in determining the ages of geomagnetic reversals by dating basalt or overlying sediments beneath known anomalies is that the age determination is made on a tiny fraction of the geologically complex volume of rock that contributes to the magnetic record of the ancient field polarity. More accurate dates for the anomalies of the last several million years have resulted from correlations with radiometrically dated magnetic stratigraphy (for example, Doell and Dalrymple, 1966; Watkins and others, 1975), and in theory this should be possible for older anomalies as well.

MAGNETIC STRATIGRAPHY

All sedimentary and volcanic rocks are deposited or emplaced in the geomagnetic field, and those with suitable mineralogy record the field direction. Thus a history of field reversals may be revealed in volcanic sequences in which the course of events is double checked by superposition and by radiometric dating. Such work has been carried back about 7 m.y. (Watkins and others, 1975), but beyond that time, the method becomes difficult because of the irregularity of volcanic events and the increasing uncertainties in radiometric age determinations. Furthermore, results in volcanic sequences are not easily tied into the paleontologic scale. Accordingly, there has been much interest in developing a magnetic stratigraphy in sedimentary rocks.

Early attention was focused mainly on red beds because of their comparatively strong remanent magnetization, but such sequences are rarely well dated by fossils. The discovery that marine biogenic oozes yield reliable magnetic results (Opdyke and others, 1966; Opdyke, 1972) stimulated interest in indurated pelagic and hemipelagic limestones and marls, which have the virtues of more continuous sedimentation and abundant datable fossils. However, the study of these weakly magnetized rocks remained unfeasible in many cases until the development of more rapid and sensitive magnetometers, such as the digital spinners and cryogenic instruments used in this investigation.

The globally synchronous reversals of the magnetic field offer the prospect of time markers for the precise correlation of diverse events on the Earth's surface. A detailed network of correlations between (1) the sea-floor anomalies, which provide the closest approach to a continuous record of geomagnetic field polarity; (2) the radiometrically dated volcanic rock sequences on land; and (3) sedimentary sections which contain the standard biostratigraphic zones is essential to further progress along these lines. A number of studies of magnetic stratigraphy in sedimentary sections have been published recently (Opdyke and others, 1974; Theyer and Hammond, 1974; Shive and Frerichs, 1974). Previous reversal time scales for the Late Cretaceous and Paleocene have been pieced together from a number of partial sections (van Hinte, 1972, 1976; Watkins, 1975; Keating and others, 1975; Thierstein, 1977). The Gubbio section provides the first complete biostratigraphic and magnetic record from a single locality.

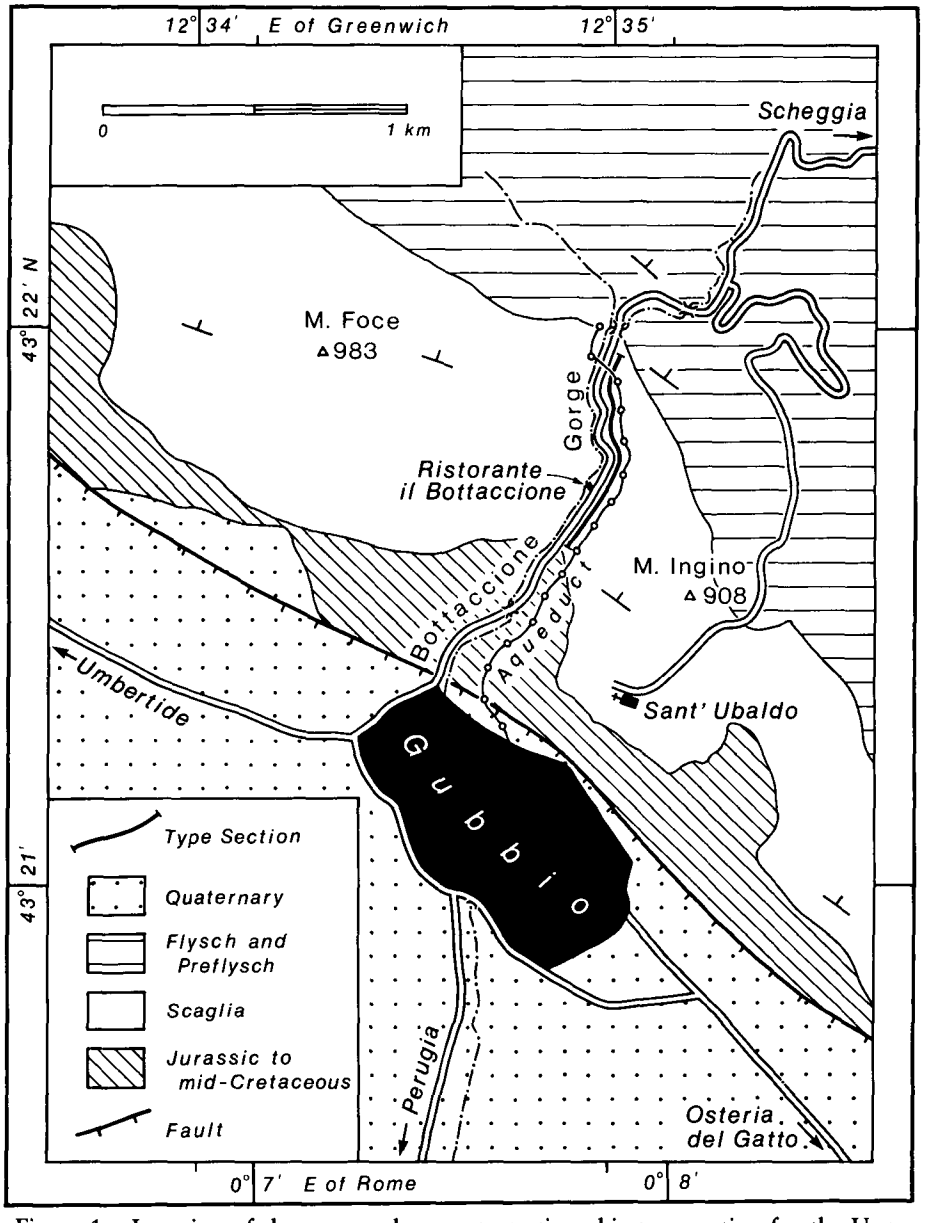


Figure 1. Location of the proposed magnetostratigraphic type section for the Upper Cretaceous and Paleocene. The geology is simplified after Barnaba (1958).

THE GUBBIO SECTION

In the course of the sedimentological, paleontological, and paleomagnetic investigations described in the four preceding papers, we have delineated the sequence of magnetic polarity zones in the Gubbio sec-

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tion and have tied this sequence directly to the standard planktonic foraminiferal zonation.

The advantages offered by the Gubbio section for this kind of study are (1) nearly complete exposure in fresh roadcuts; (2) relatively constant rate and type of sedimentation, yielding limestones which provide both a magnetic record and the full sequence of known planktic foraminiferal zones — this stratigraphic continuity apparently extends across the Maastrichtian-Paleocene boundary, which in much of the world is marked by a hiatus; (3) abundant foraminifera; (4) remanent magnetization that is stable and well clustered after AF or thermal removal of secondary components, and which, although weak, can be measured routinely on digital spinner magnetometers or cryogenic instruments; (5) good access; and (6) a well-indurated lithology which is easily sampled with a portable drill.

The principal disadvantage is that the foraminifera cannot usually be separated from the matrix and must be identified in thin section, while nannoplankton are not well preserved, and other fossils are rare. A second disadvantage is the presence of some small faults, most of which have displacements of less than 1 m, but a few of which may have somewhat larger offsets. However, the continuity of faunal zones and the close match between the reversal sequence at Gubbio and the marine magnetic anomaly sequence (Lowrie and Alvarez, this issue, Fig. 2) confirm the absence of major structural disturbances.

PROPOSED MAGNETOSTRATIGRAPHIC TYPE SECTION

Inasmuch as magnetic stratigraphy will have to be patched together from many scattered sections, as has biostratigraphy, it is important to avoid procedural pitfalls that could lead to the kind of ambiguity and fruitless argument that have troubled classical stratigraphy. We thus, in general, adhere to the recommendations of the Subcommission of Magnetostratigraphic Nomenclature (1973). The most essential of their recommendations is the conceptual separation of concrete, measurable and mappable rock units from inferred units of time. We feel that it is equally important that magnetostratigraphic and chronostratigraphic entities or concepts be tied to type sections, where they may subsequently be tested, emended, and elaborated.

Because of the apparently complete reversal sequence recorded in the Scaglia Rossa at Gubbio, we propose that this be taken as the type section for the Late Cretaceous–Paleocene geomagnetic reversal sequence. In accordance with the procedure for establishing a type section given in the Code of Stratigraphic Nomenclature (Am. Commission on Stratigraphic Nomenclature, 1961), we note that the section is located at 43°21.6–22.0′N, 12°34.9′E of Greenwich (0°7.8′E of Rome, Monte Mario) on Italian topographic map sheet 116-III-SE "Gubbio" (1:25,000). The outcrops are on the southeast side of the Gubbio-Scheggia road, between kilometre posts 10 and 11 (km 41-42 in the old numbering shown on the topographic map), below the mediaeval aqueduct known as the "Bottaccione." A location map is given in Figure 1.

The physical magnetostratigraphic units defined at Gubbio are sequences of beds characterized by a common polarity. These are polarity zones; we have recognized 23 in the Gubbio section. For purposes of reference they require identifying labels. The lowest of the zones, which appears to represent the Long Normal episode of the Cretaceous, is here identified as the "Gubbio Long Normal Zone." The succeeding zones are designated by letters with the prefix "Gubbio," in ascending alphabetical order; normal zones are distinguished by "+" and reversed zones by "−." In accordance with the hierarchy of units recommended by the Subcommission on Magnetostratigraphic Nomenclature (1973), we have chosen to regard some of the thinnest polarity units as polarity subzones within thicker polarity zones. Thus subzones Gubbio D1+, D2−, and D3+ make up zone Gubbio D+. This procedure is analogous to standard practice in lithostratigraphy, where the section must be completely covered by a sequence of formations, while some (but not necessarily all) of the formations may be divided into members.

The zonal designations are shown in our Figure 2; in paper III, Figures 1 and 2; and in paper IV, Figure 3. The positions of magnetic zonal boundaries relative to foraminiferal zones and to various proposed absolute chronologies are also given in Figure 2. Metre numbers have been painted on the outcrop, and more permanent markers for polarity-zone boundaries have been cemented into the rock.

We infer that each polarity zone recognized in the Gubbio section was formed during a period of corresponding normal or reversed geomagnetic polarity. These abstract units of Earth history we term "polarity chrons" in adaptation to Hedberg (1976, p. 69), rather than "polarity intervals" as suggested by the Subcommission (1973), inasmuch as "chron" refers clearly to time, whereas "interval" is ambiguous, as pointed out to us by Hedberg. These units of time may be designated with the same labels as their corresponding zones in the rock record; thus one may refer to the "Gubbio G− Polarity Chron," a unit of time that included the very latest Cretaceous and the very earliest Paleocene.

Correlations between the sequence of polarity zones at Gubbio and the sequence of marine magnetic anomalies have been presented in the preceding papers. Lowrie and Alvarez (paper III) assumed the Gubbio Long Normal Zone and the "Cretaceous Quiet Zone" of the sea floor to be common expressions of the Cretaceous Long Normal polarity chron and found an excellent match of successive longer and shorter polarity zones at Gubbio with the sequence of marine magnetic anomalies. In this match, the Cretaceous-Tertiary boundary falls into polarity zone Gubbio G− and between anomalies 29 and 30. Independently, Roggenthen and Napoleone (paper IV) correlated the long zone Gubbio M− with the broad negative anomaly between 26 and 27 and used the paleontologic data available from DSDP–JOIDES (paper II, Table 1) as tie points (paper IV, Fig. 4). Matching polarity zones with sea-floor anomalies upward and downward yielded a correlation for the top of the Cretaceous identical to that arrived at by Lowrie and Alvarez (paper III) and identified anomaly 26 as belonging to the late Paleocene *pseudomenardii* zone. These correlations agree with the age assignments of marine magnetic anomalies by Sclater and others (1974).

PALEONTOLOGICAL CALIBRATION

In Figure 2, the stratigraphic positions of the reversal boundaries in the Gubbio section are compared with the stage and foraminiferal zone boundaries determined by Premoli Silva (this issue). The following observations can be made:

1. Although the base of the Gubbio (Cretaceous) Long Normal Zone is not dated by this study, it is at least as old as Cenomanian.
2. The Gubbio Long Normal Zone ends very close to the Santonian-Campanian boundary, with the onset of the Gubbio A− Zone.
3. The Campanian-Maastrichtian boundary falls near the top of the relatively long zone Gubbio B+.
4. All Late Cretaceous reversed polarity zones except Gubbio A− are of Maastrichtian age.
5. In contrast with the Campanian, the Paleocene and Maastrichtian are generally

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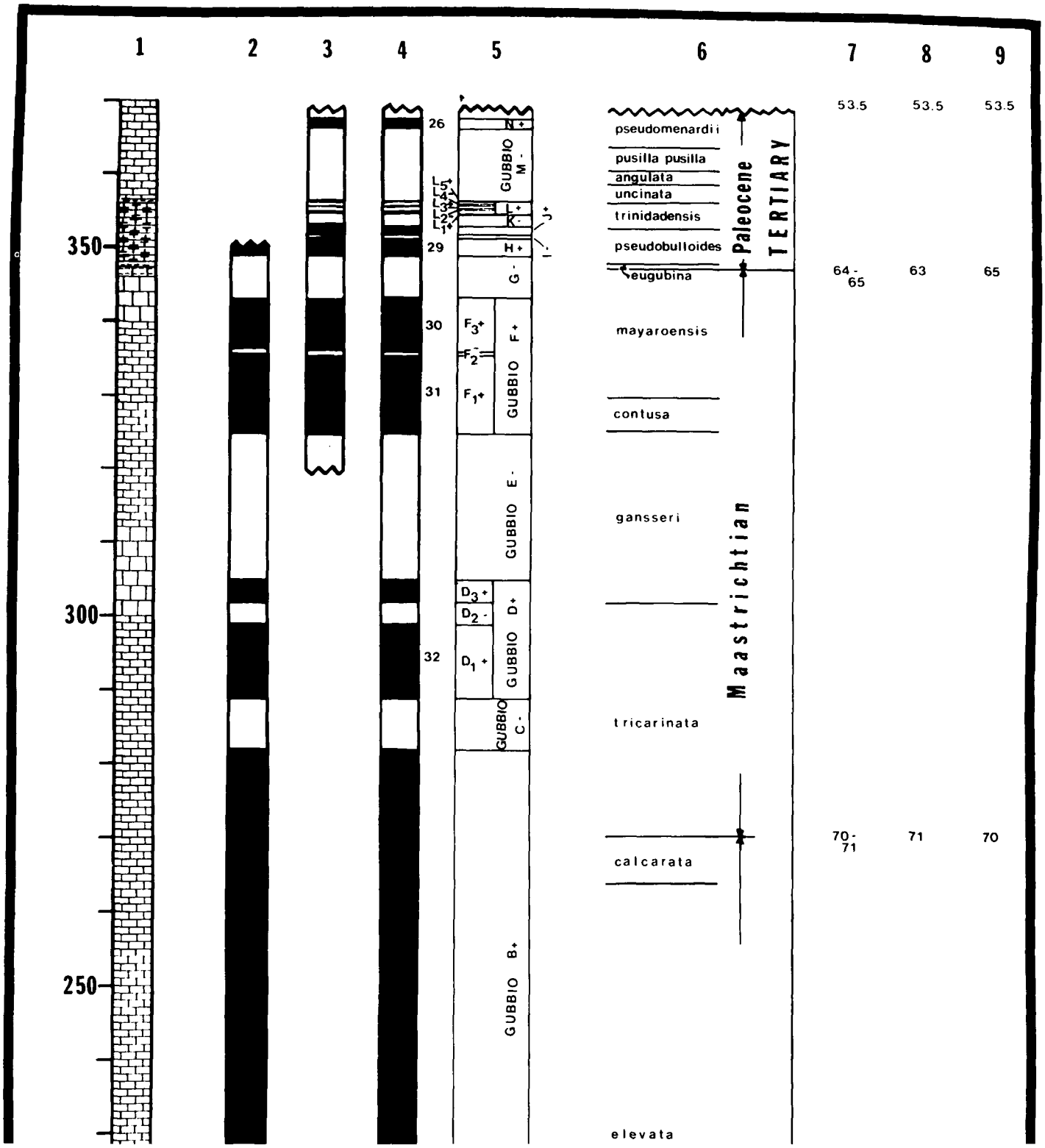


Figure 2. Results of stratigraphic studies of the Gubbio section. Column 1: measured section in metres with lithology (Arthur and Fischer, paper I). Column 2: magnetic results of Lowrie and Alvarez (paper III); black is normal polarity, white is reversed. Column 3: magnetic results of Roggenthen and Napoleone (paper IV). Column 4: combined magnetic results (numbers to the right of the column show correlations with the marine magnetic anomaly sequence). Column 5: designation of polarity zones; "+" indicates normal polarity, "−" indicates reversed polarity; zones Gubbio D+, F+, and L+ have been divided into subzones. Column 6: planktic foraminiferal zones from Premoli Silva (paper II), with chronostratigraphic units on the right. Columns 7 through 9: absolute age calibration according to three proposed time scales. 7: Obradovich and Cobban (1975). 8: Bukry and Douglas (unpub. data; 1974, personal commun.). 9: van Hinte (1972). Paleocene-Eocene boundary age from Berggren (1972) in all cases. Exact metre levels of boundaries between magnetic polarity zones and biostratigraphic zones in the measured section are reported in the preceding papers.

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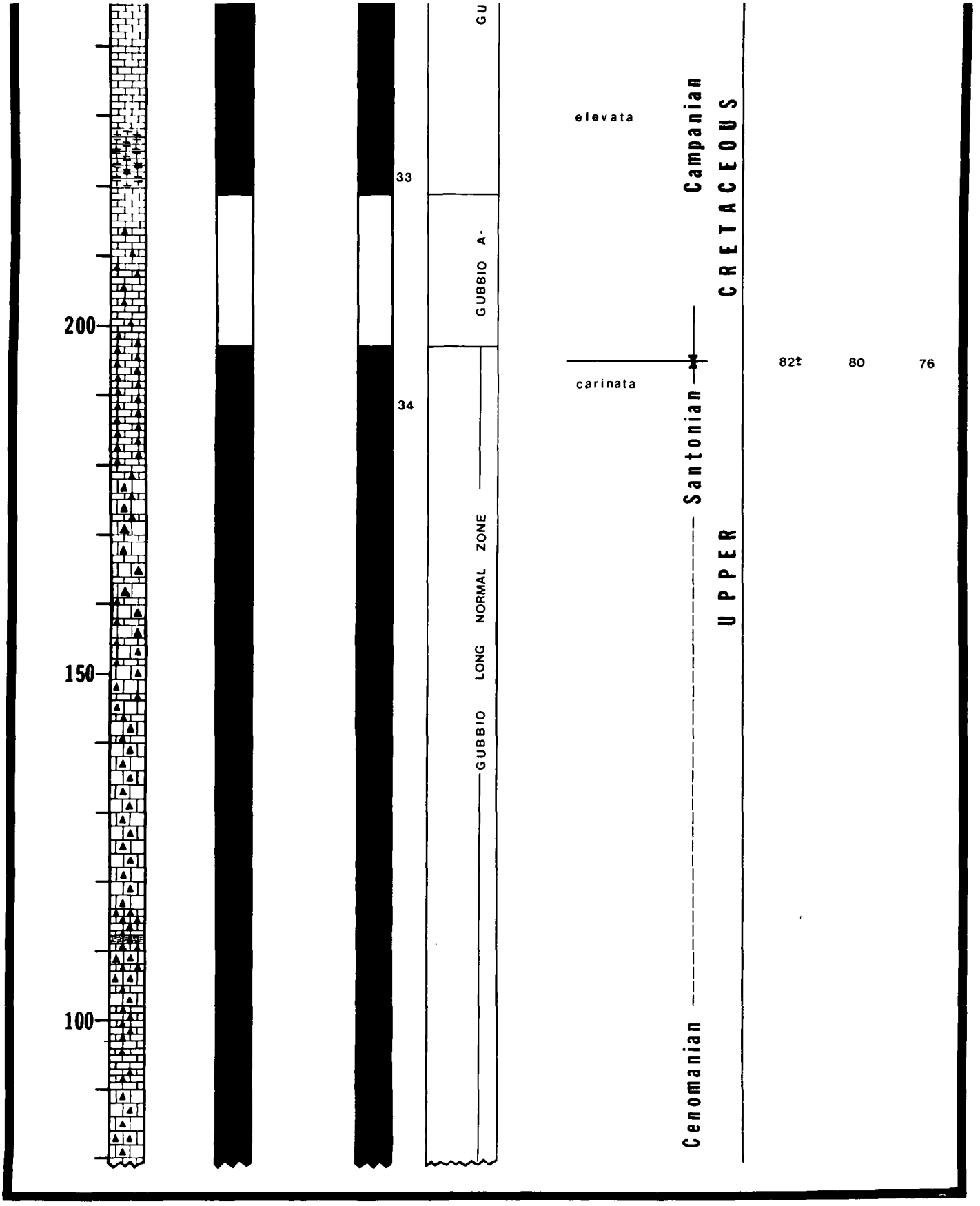


Figure 2. (*Continued*).

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characterized by polarity zones that represent short time intervals.

6. The Cretaceous-Tertiary boundary falls nearly at the top of polarity zone Gubbio G−.

7. The section studied does not reach the Paleocene-Eocene boundary, which must occur slightly above the Gubbio N+ zone.

CALIBRATION BY ABSOLUTE AGE

Calibration of geomagnetic polarity history in terms of years remains somewhat uncertain because of inaccuracies in radiometric age determinations and in correlations between radiometrically dated rocks and those rocks that contain fossils. The present state of knowledge is illustrated by the dates assigned to various biostratigraphic boundaries in three contemporary time scales, as shown in Figure 2 (cols. 7 through 9).

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